7

EXPANSIONS AND FAITHFUL INTERPRETATIONS

In this chapter we introduce the concepts of expansion and faithful interpretation of a geometric theory, establish a number of general results about them and apply them in the context of theories of presheaf type.

In section 7.1, we investigate expansions of geometric theories from the point of view of the geometric morphisms that they induce between the respective classifying toposes. In particular, we introduce the notion of localic (resp. hyperconnected) expansion, and show that it naturally corresponds to the notion of localic (resp. hyperconnected) geometric morphism at the level of classifying toposes; as a result, we obtain a syntactic description of the hyperconnected-localic factorization of a geometric morphism. Next, we address the problem of expanding a given geometric theory $\mathbb{T}$ to a theory classified by the topos $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$, and describe a general method for constructing such expansions (which we call *presheaf completions*). Along the way, we establish a new criterion for a geometric theory to be of presheaf type.

In section 7.2, we investigate to what extent the satisfaction of the conditions of Theorem 6.3.1 is preserved by faithful interpretations of geometric theories. Notable examples of faithful interpretations are given by quotients and injectivizations (the latter being theories obtained from a given geometric theory by adding, for each sort of its signature, a binary predicate which is provably a complement of the equality relation on that sort). We focus on the case of injectivizations – the case of quotients being treated in section 8.2 – providing sufficient conditions for the injectivization $\mathbb{T}_m$ of a geometric theory $\mathbb{T}$ to be of presheaf type if $\mathbb{T}$ is so. To this end, we carry out a general analysis of the relationship between finitely presentable and finitely generated models of a given geometric theory.

7.1 **Expansions of geometric theories**

7.1.1 *General theory*

In this section we introduce the syntactic notion of expansion of a geometric theory, and show that it corresponds in a natural way to that of geometric morphism between the respective classifying toposes. Further, we identify classes of expansions of theories whose corresponding geometric morphisms are localic (resp. hyperconnected). This development has been inspired by section 6.2 of [41], which contains a brief informal discussion of this topic.

Theories, Sites, Toposes. Olivia Caramello.

Definition 7.1.1 Let $\mathbb{T}$ be a geometric theory over a signature Σ.

(a) A geometric *expansion* of $\mathbb{T}$ is a geometric theory obtained from $\mathbb{T}$ by adding sorts, relation or function symbols to the signature Σ and geometric axioms over the resulting extended signature; equivalently, a geometric expansion of $\mathbb{T}$ is a geometric theory $\mathbb{T}'$ over a signature containing Σ such that every axiom of $\mathbb{T}$, regarded as a geometric sequent over the signature of $\mathbb{T}'$, is provable in $\mathbb{T}'$.

(b) A geometric expansion $\mathbb{T}'$ of $\mathbb{T}$ is said to be *localic* if no new sorts are added to Σ to obtain the signature of $\mathbb{T}'$.

(c) A geometric expansion $\mathbb{T}'$ of $\mathbb{T}$ is said to be *hyperconnected* if every geometric formula over Σ' in a context entirely consisting of sorts of Σ is provably equivalent in $\mathbb{T}'$ to a geometric formula in the same context over Σ, and for any geometric sequent σ over Σ, σ is provable in $\mathbb{T}'$ if and only if it is provable in $\mathbb{T}$.

If $\mathbb{T}'$ is an expansion of a geometric theory $\mathbb{T}$ then there is a canonical morphism of sites $(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \to (\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$ inducing a geometric morphism $p_{\mathbb{T}}^{\mathbb{T}'} : \mathbf{Set}[\mathbb{T}'] \to \mathbf{Set}[\mathbb{T}]$.

We know from Theorem 4.2.13 that the inclusion part of the surjection-inclusion factorization of the geometric morphism $p_{\mathbb{T}}^{\mathbb{T}'} : \mathbf{Set}[\mathbb{T}'] \to \mathbf{Set}[\mathbb{T}]$ can be identified with the classifying topos of the quotient of $\mathbb{T}$ consisting of all the geometric sequents over Σ which are provable in the theory $\mathbb{T}'$.

Recall (cf. for instance section A4.6 of [55]) that a geometric morphism $f : \mathcal{F} \to \mathcal{E}$ is *localic* if every object of $\mathcal{F}$ is a subquotient (i.e. a quotient of a subobject) of an object of the form $f^*(a)$, where a is an object of $\mathcal{E}$; the morphism f is *hyperconnected* if f^* is full and faithful and its image is closed under subobjects in $\mathcal{F}$.

The following technical lemma will be useful in the sequel.

Lemma 7.1.2 *Let $f : \mathcal{F} \to \mathcal{E}$ be a geometric morphism between Grothendieck toposes and $\mathcal{C}$ a small, full subcategory of $\mathcal{E}$ which is separating for $\mathcal{E}$. Suppose that the following conditions are satisfied:*

(i) *$\mathcal{C}$ is closed in $\mathcal{E}$ under finite products.*

(ii) *f satisfies the property that the restriction $f^*|_{\mathcal{C}} : \mathcal{C} \to \mathcal{F}$ of f^* to $\mathcal{C}$ is full and faithful.*

(iii) *For any family of arrows $\mathcal{T}$ in $\mathcal{C}$ with common codomain, if the image of $\mathcal{T}$ under f^* is epimorphic in $\mathcal{F}$ then $\mathcal{T}$ is epimorphic in $\mathcal{E}$.*

(iv) *Every subobject in $\mathcal{F}$ of an object of the form $f^*(c)$, where c is an object of $\mathcal{C}$, is, up to isomorphism, of the form $f^*(m)$ (where m is a subobject of c in $\mathcal{E}$).*

Then f is hyperconnected.

Proof We have to prove that, under the specified hypotheses, $f^* : \mathcal{E} \to \mathcal{F}$ is full and faithful and its image is closed under subobjects in $\mathcal{F}$.

To prove that f^* is faithful, we have to verify that for any arrows $h, k : u \to v$ in $\mathcal{E}$, $f^*(h) = f^*(k)$ implies $h = k$. Since $\mathcal{C}$ is separating for $\mathcal{E}$, we can suppose without loss of generality that u lies in $\mathcal{C}$. Now, $h = k$ if and only if the equalizer $z : w \rightarrowtail u$ of h and k is an isomorphism. The family of arrows from objects of $\mathcal{C}$ to w is epimorphic, so, as $f^*(z)$ is an isomorphism (since $f^*(h) = f^*(k)$), the family formed by the composition of these arrows with z is epimorphic on u; indeed, the members of this latter family all lie in $\mathcal{C}$ and the image of this family under f^* is epimorphic. It follows that z is an epimorphism, equivalently an isomorphism, that is, $h = k$, as required.

To prove the fullness of f^*, we have to verify that for any objects a and b of $\mathcal{E}$ and any arrow $s : f^*(a) \to f^*(b)$ in $\mathcal{F}$, there exists a (unique) arrow $t : a \to b$ in $\mathcal{E}$ such that $f^*(t) = s$. As $\mathcal{C}$ is separating for $\mathcal{E}$, there exist epimorphic families $\{ f_i : c_i \to b \mid i \in I\}$ and $\{g_j : d_j \to a \mid j \in J\}$ in $\mathcal{E}$ consisting of arrows whose domains lie in $\mathcal{C}$.

For any $i \in I$ and $j \in J$, consider the following pullback square:

$$\begin{tikzcd} r_{i,j} \arrow[r, "p_i"] \arrow[d, "q_j"] & f^*(c_i) \arrow[d, "f^*(f_i)"] \\ f^*(d_j) \arrow[r, "s \circ f^*(g_j)"] & f^*(b). \end{tikzcd}$$

By the universal property of pullbacks, the arrow $< p_i, q_j >: r_{i,j} \to f^*(c_i) \times f^*(d_j) \cong f^*(c_i \times d_j)$ is a monomorphism. Since $c_i \times d_j$ lies in $\mathcal{C}$ by our hypothesis, $r_{i,j}$ lies in the image of f^* and hence can be covered by arrows $h_k^{i,j} : f^*(e_k^{i,j}) \to r_{i,j}$ (for $k \in K_{i,j}$) where each object $e_k^{i,j}$ lies in $\mathcal{C}$. Consider the arrows $p_i \circ h_k^{i,j} : f^*(e_k^{i,j}) \to f^*(c_i)$ and $q_j \circ h_k^{i,j} : f^*(e_k^{i,j}) \to f^*(d_j)$. By our hypotheses, there exist arrows $l_k^{i,j} : e_k^{i,j} \to c_i$ and $m_k^{i,j} : e_k^{i,j} \to d_j$ in $\mathcal{C}$ such that $f^*(l_k^{i,j}) = p_i \circ h_k^{i,j}$ and $f^*(m_k^{i,j}) = q_j \circ h_k^{i,j}$. Since the family $\{f^*(g_j \circ m_k^{i,j}) \mid i \in I, j \in J, k \in K_{i,j}\}$ is epimorphic in $\mathcal{F}$, the faithfulness of f^* ensures that the family $\{g_j \circ m_k^{i,j} : e_k^{i,j} \to a \mid i \in I, j \in J, k \in K_{i,j}\}$ is epimorphic in $\mathcal{E}$. Defining an arrow $t : a \to b$ in $\mathcal{E}$ is thus equivalent to specifying, for each $i \in I$, $j \in J$ and $k \in K_{i,j}$, an arrow $v_k^{i,j} : e_k^{i,j} \to b$ in such a way that the compatibility conditions of Corollary 5.1.7 are satisfied. Take $v_k^{i,j}$ equal to $f_i \circ l_k^{i,j}$. The compatibility conditions hold since the images of them under the functor f^* satisfy them and, as we have proved above, f^* is faithful. Thus we have a unique arrow $t : a \to b$ in $\mathcal{E}$ such that $t \circ g_j \circ m_k^{i,j} = f_i \circ l_k^{i,j}$ (for all $i \in I$, $j \in J$, $k \in K_{i,j}$). The fact that $f^*(t) = s$ follows at once.

To conclude the proof of the lemma, it remains to show that the image of f^* is closed under subobjects in $\mathcal{F}$. Let a be an object of $\mathcal{E}$ and $k : z \rightarrowtail f^*(a)$ a subobject in $\mathcal{F}$. Since $\mathcal{C}$ is separating for $\mathcal{E}$, the canonical arrow from the coproduct of all the objects in $\mathcal{C}$ to a is an epimorphism, call it y. Pulling back $f^*(y)$ along the monomorphism k, we obtain a monomorphism $u : z' \to \mathrm{dom}(f^*(y))$ and an epimorphism $e : z' \to z$. Now, by our hypotheses, the pullbacks of u along the coproduct arrows belong to the image of f^*; from the fact that f^* preserves coproducts it thus follows that u itself belongs to the image of f^*, it being of the form $f^*(m)$ where m is a coproduct of subobjects in $\mathcal{E}$ of objects of $\mathcal{C}$. Now, since e is an epimorphism, z is isomorphic to the coequalizer of its kernel pair $r : w \rightarrowtail f^*(b) \times f^*(b) \cong f^*(b \times b)$ where $b = \mathrm{dom}(m)$. Using the coproduct representation of m and the fact that $\mathcal{C}$ is closed under finite products in $\mathcal{E}$, one can prove, by considering the pullbacks of r along the images under f^* of the induced coproduct arrows to $b \times b$, that r lies, up to isomorphism, in the image of f^*; therefore, as f^* preserves coequalizers, k is isomorphic to a subobject in the image of f^*, as required. $\square$

Theorem 7.1.3 *Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathbb{T}'$ a geometric expansion of $\mathbb{T}$ over a signature Σ'. If $\mathbb{T}'$ is a hyperconnected (resp. localic) expansion of $\mathbb{T}$ then $p_{\mathbb{T}}^{\mathbb{T}'} : \mathbf{Set}[\mathbb{T}'] \to \mathbf{Set}[\mathbb{T}]$ is a hyperconnected (resp. localic) geometric morphism. In particular, if we identify $\mathbb{T}'$ with its (Morita-equivalent) expansion obtained by adding, for each finite string of sorts $A_1 \cdots A_n$ of Σ and each geometric formula $\phi(\vec{x}) = \phi(x_1^{A_1}, \ldots, x_n^{A_n})$ over Σ' (considered up to provable equivalence), a relation symbol $R_{\phi(\vec{x})} \rightarrowtail A_1 \cdots A_n$ and the axiom $(R_{\phi(\vec{x})} \dashv\vdash_{\vec{x}} \phi)$, the hyperconnected-localic factorization of the geometric morphism $p_{\mathbb{T}}^{\mathbb{T}'} : \mathbf{Set}[\mathbb{T}'] \to \mathbf{Set}[\mathbb{T}]$ is given by $p_{\mathbb{T}}^{\mathbb{T}''} \circ p_{\mathbb{T}''}^{\mathbb{T}'}$, where $\mathbb{T}''$ is the intermediate expansion of $\mathbb{T}$ obtained by adding to the signature Σ of $\mathbb{T}$ no new sorts and the relation symbols $R_{\phi(\vec{x})}$ and taking as axioms all the sequents over this extended signature which are provable in $\mathbb{T}'$ when each occurrence of $R_{\phi(\vec{x})}$ is replaced by the formula $\phi(\vec{x})$ which defines it.*

Proof Suppose that $\mathbb{T}'$ is a localic expansion of $\mathbb{T}$. We have to prove that every object of $\mathbf{Set}[\mathbb{T}']$ is a quotient of a subobject of an object of the form $(p_{\mathbb{T}}^{\mathbb{T}'})^*(H)$ where H is an object of $\mathbf{Set}[\mathbb{T}]$. Since the signature of $\mathbb{T}'$ does not contain any sorts not already present in the signature of $\mathbb{T}$ and the images of the objects of the category $\mathcal{C}_{\mathbb{T}'}$ under the Yoneda embedding $y_{\mathbb{T}'} : \mathcal{C}_{\mathbb{T}'} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$ form a separating (essentially small) family of objects for the topos $\mathbf{Set}[\mathbb{T}']$, we can conclude that the subobjects of the objects of the form $(p_{\mathbb{T}}^{\mathbb{T}'})^*(y_{\mathbb{T}}(\{\vec{x} \,.\, \top\})) \cong y_{\mathbb{T}'}(\{\vec{x} \,.\, \top\})$, where $y_{\mathbb{T}}$ is the Yoneda embedding $\mathcal{C}_{\mathbb{T}} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$, form a separating, essentially small family of objects for the topos $\mathbf{Set}[\mathbb{T}']$. Therefore any object of $\mathbf{Set}[\mathbb{T}']$ is a quotient of a coproduct of subobjects of objects of the form $(p_{\mathbb{T}}^{\mathbb{T}'})^*(y_{\mathbb{T}}(\{\vec{x} \,.\, \top\}))$; but the fact that $(p_{\mathbb{T}}^{\mathbb{T}'})^*$ preserves coproducts implies that any such coproduct is a subobject of an object in the image of $(p_{\mathbb{T}}^{\mathbb{T}'})^*$.

Suppose instead that $\mathbb{T}'$ is a hyperconnected expansion of $\mathbb{T}$. We have to prove that $(p_{\mathbb{T}}^{\mathbb{T}'})^*$ is hyperconnected. It suffices to apply Lemma 7.1.2 by taking $\mathcal{E}$ equal to $\mathbf{Set}[\mathbb{T}]$ and $\mathcal{C}$ equal to (a skeleton of) the syntactic category $\mathcal{C}_{\mathbb{T}}$; the fact that the hypotheses of the lemma are satisfied follows immediately from the fact that $\mathbb{T}'$ is hyperconnected over $\mathbb{T}$. $\square$

Remark 7.1.4 Let $\mathbb{T}'$ be a geometric expansion of a geometric theory $\mathbb{T}$. It is immediate to see that the property of this expansion to be hyperconnected is not only sufficient for the geometric morphism $p_{\mathbb{T}}^{\mathbb{T}'}$ to be hyperconnected (by Theorem 7.1.3), but also necessary. On the other hand, it is not necessary for $p_{\mathbb{T}}^{\mathbb{T}'}$ to be localic that $\mathbb{T}'$ be a localic expansion of $\mathbb{T}$; indeed, there are non-propositional theories classified by localic toposes (see, for instance, sections 3.4 and 3.5 of [74]).

The following theorem provides a converse to Theorem 7.1.3.

Theorem 7.1.5 *Let $p : \mathcal{E} \to \mathbf{Set}[\mathbb{T}]$ be a geometric morphism to the classifying topos of a geometric theory $\mathbb{T}$. Then p is, up to isomorphism, of the form $p_{\mathbb{T}}^{\mathbb{T}'}$ for some geometric expansion $\mathbb{T}'$ of $\mathbb{T}$. If p is hyperconnected (resp. localic) then we can take $\mathbb{T}'$ to be a hyperconnected (resp. localic) expansion of $\mathbb{T}$.*

Proof Choose a triple $\mathcal{T} = (\mathcal{C}_{\text{ob}}, \mathcal{C}_{\text{arr}}, \mathcal{C}_{\text{rel}})$ consisting of a set $\mathcal{C}_{\text{ob}}$ of objects of $\mathcal{E}$, a set $\mathcal{C}_{\text{arr}}$ of arrows in $\mathcal{E}$ from finite products of objects of $\mathcal{C}_{\text{ob}}$ to objects of $\mathcal{C}_{\text{ob}}$ and a set $\mathcal{C}_{\text{rel}}$ of subobjects of finite products of objects of $\mathcal{C}_{\text{ob}}$ with the property that the family of objects of $\mathcal{E}$ which can be built up from objects in $\mathcal{C}_{\text{ob}}$, arrows in $\mathcal{C}_{\text{arr}}$ and subobjects in $\mathcal{C}_{\text{rel}}$ by using geometric logic constructions is separating for $\mathcal{E}$. By definition of a Grothendieck topos, such a triple always exists. Let us define an expansion $\mathbb{T}_{\mathcal{T}}$ of $\mathbb{T}$ as follows. The signature $\Sigma_{\mathbb{T}_{\mathcal{T}}}$ of $\mathbb{T}_{\mathcal{T}}$ is obtained by adding to the signature of $\mathbb{T}$ one sort $\ulcorner c \urcorner$ for each object c in $\mathcal{C}_{\text{ob}}$ (or, more economically, for each c which is not, up to isomorphism, of the form $f^*(H)$ for some object H of $\mathcal{C}_{\mathbb{T}} \hookrightarrow \mathbf{Set}[\mathbb{T}]$), one function symbol $\ulcorner f \urcorner$ for each arrow f in $\mathcal{C}_{\text{arr}}$ (or, more economically, for each f whose domain or codomain is not, up to isomorphism, of the form $f^*(H)$ for some object H of $\mathcal{C}_{\mathbb{T}} \hookrightarrow \mathbf{Set}[\mathbb{T}]$), whose sorts are the obvious ones, one relation symbol for each subobject in $\mathcal{C}_{\text{rel}}$ (whose sorts are the obvious ones, those corresponding to an object of the form $f^*(\{\vec{x} \,.\, \phi\})$ being the sorts of the variables in $\vec{x}$) and an additional relation symbol $\ulcorner R \urcorner$ for any subobject $R \rightarrowtail c_1 \times \cdots \times c_n$ in $\mathcal{E}$ where $c_1, \ldots, c_n$ are objects in $\mathcal{C}_{\text{ob}}$ (or, more economically, for each R which cannot be obtained, up to isomorphism, from the data in $\mathcal{T}$ by means of geometric logic constructions), whose sorts are the obvious ones.

Consider the tautological $\Sigma_{\mathbb{T}_{\mathcal{T}}}$-structure M in $\mathcal{E}$ obtained by interpreting each sort $\ulcorner c \urcorner$ by the corresponding object c (and similarly for the function and relation symbols added to the signature of $\mathbb{T}$), and each sort A of the signature of $\mathbb{T}$ by the object $f^*(\{x^A \,.\, \top\})$ (and similarly for the function and relation symbols of the signature of $\mathbb{T}$). Define $\mathbb{T}_{\mathcal{T}}$ to be the theory $\text{Th}(M)$ of M over the signature $\Sigma_{\mathbb{T}_{\mathcal{T}}}$ (in the sense of Definition 4.2.12). The theory $\mathbb{T}' := \mathbb{T}_{\mathcal{T}}$ satisfies the conditions of Theorem 2.1.29; the validity of the first two conditions is obvious, while the validity of the third follows from the fact that any subobject of a product of objects in $\mathcal{C}_{\text{ob}}$ is definable in $\mathbb{T}'$ and the model M is conservative for $\mathbb{T}'$, whence the formula defining the graph of the given arrow is $\mathbb{T}'$-provably functional from the formula defining the domain to the formula defining the codomain. Therefore $\mathbb{T}'$ is classified by the topos $\mathcal{E}$ with universal model M. Since $\mathbb{T}'$ is an expansion of $\mathbb{T}$, this proves the first part of the theorem.

Suppose now that f is localic. We can define a triple $\mathcal{T} = (\mathcal{C}_{\mathrm{ob}}, \mathcal{C}_{\mathrm{arr}}, \mathcal{C}_{\mathrm{rel}})$ satisfying the conditions specified above as follows: we set $\mathcal{C}_{\mathrm{ob}}$ equal to the set of objects of the form $f^*(H)$ where H is an object of $\mathcal{C}_{\mathbb{T}} \hookrightarrow \mathbf{Set}[\mathbb{T}]$, $\mathcal{C}_{\mathrm{arr}}$ equal to the empty set and $\mathcal{C}_{\mathrm{rel}}$ equal to the set of all subobjects of (finite products of) objects in $\mathcal{C}_{\mathrm{ob}}$. Alternatively, one could take $\mathcal{C}_{\mathrm{ob}}$ to be the set of objects of the form $f^*(\{x^A . \top\})$ (where A is any sort of the signature of $\mathbb{T}$), $\mathcal{C}_{\mathrm{arr}}$ to be the set of arrows of the form $f^*([\theta])$, where $[\theta]$ is an arrow in $\mathcal{C}_{\mathbb{T}}$, and $\mathcal{C}_{\mathrm{rel}}$ to be the set of all subobjects of (finite products of) objects in $\mathcal{C}_{\mathrm{ob}}$. In either case, the most economical version of the signature $\Sigma_{\mathbb{T}_{\mathcal{T}}}$ of $\mathbb{T}_{\mathcal{T}}$ will contain no new sorts with respect to the signature of $\mathbb{T}$ and hence $\mathbb{T}_{\mathcal{T}}$ will be a localic expansion of $\mathbb{T}$.

Suppose instead that f is hyperconnected. Given any set $\mathcal{K}$ of objects of $\mathcal{E}$ which, together with the objects of the form $f^*(H)$ (where H is an object of $\mathcal{C}_{\mathbb{T}} \hookrightarrow \mathbf{Set}[\mathbb{T}]$), is separating for $\mathcal{E}$, we can define a triple $\mathcal{T} = (\mathcal{C}_{\mathrm{ob}}, \mathcal{C}_{\mathrm{arr}}, \mathcal{C}_{\mathrm{rel}})$ satisfying the required conditions by setting $\mathcal{C}_{\mathrm{ob}} = \mathcal{K}$, $\mathcal{C}_{\mathrm{arr}} = \emptyset$ and $\mathcal{C}_{\mathrm{rel}} = \emptyset$. Since f is hyperconnected, the image of f^* is closed under subobjects; hence every geometric formula over the signature of $\mathbb{T}'$ in a context entirely consisting of sorts of Σ is provably equivalent in $\mathbb{T}'$ to a geometric formula in the same context over the signature of $\mathbb{T}$. On the other hand, any geometric sequent over the signature of $\mathbb{T}$ is provable in $\mathbb{T}'$ if and only if it is provable in $f^*(U_{\mathbb{T}})$, where $U_{\mathbb{T}}$ is 'the' universal model of $\mathbb{T}$ lying in $\mathbf{Set}[\mathbb{T}]$ (since f^* is full and faithful), i.e. if and only if it is provable in $\mathbb{T}$. Hence $\mathbb{T}'$ is a hyperconnected expansion of $\mathbb{T}$, as required. $\square$

7.1.2 *Another criterion for a theory to be of presheaf type*

The following criterion for a geometric theory to be of presheaf type will be useful in connection with the problem of expanding a given geometric theory to a theory of presheaf type addressed in section 7.1.3.

Theorem 7.1.6 *Let $\mathbb{T}$ be a geometric theory over a signature Σ. Then $\mathbb{T}$ is of presheaf type if and only if the following conditions are satisfied:*

(i) *Every finitely presentable model is presented by a geometric formula over Σ.*

(ii) *Every property of finite tuples of elements of a finitely presentable $\mathbb{T}$-model which is preserved by $\mathbb{T}$-model homomorphisms is definable (in finitely presentable $\mathbb{T}$-models) by a geometric formula over Σ.*

(iii) *The finitely presentable $\mathbb{T}$-models are jointly conservative for $\mathbb{T}$.*

Proof The fact that every theory of presheaf type satisfies the given conditions has been established in section 6.1 (cf. Theorems 6.1.4 and 6.1.13 and Remark 6.1.2(c)). It thus remains to prove the 'if' part of the theorem.

Consider the Σ-structure U in the topos $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ given by the forgetful functors at each sort. Clearly, U is a $\mathbb{T}$-model. To deduce our thesis, we shall verify that $\mathbb{T}$ satisfies the three conditions of Theorem 2.1.29 with respect to the model U.

Since every finitely presentable $\mathbb{T}$-model is presented by a geometric formula over Σ, the first condition of the theorem is satisfied; indeed, every representable functor $\mathrm{Hom}_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})}(c, -)$ is isomorphic to the interpretation of a geometric formula $\phi(\vec{x})$ in the model U (take as $\phi(\vec{x})$ any formula presenting c). The second condition of the theorem follows immediately from the fact that the finitely presentable $\mathbb{T}$-models are jointly conservative for $\mathbb{T}$. It remains to show that the third condition of Theorem 2.1.29 is satisfied. To this end, we observe that for any geometric formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ over the signature of $\mathbb{T}$, the graph of any arrow $[[\vec{x} \,.\, \phi]]_U \to [[\vec{y} \,.\, \psi]]_U$ in the topos $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ is a subobject of the product $[[\vec{x} \,.\, \phi]]_U \times [[\vec{y} \,.\, \psi]]_U$ in $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ and hence it defines a property of tuples of elements of finitely presentable models of $\mathbb{T}$ which is preserved by $\mathbb{T}$-model homomorphisms; so, by our assumptions, there exists a formula $\theta(\vec{x}, \vec{y})$ over Σ whose interpretation in U coincides with this subobject. Since U is conservative and this subobject is the graph of an arrow in the topos $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$, it follows that the formula $\theta(\vec{x}, \vec{y})$ is $\mathbb{T}$-provably functional from $\phi(\vec{x})$ to $\psi(\vec{y})$, as required. □

7.1.3 *Expanding a geometric theory to a theory of presheaf type*

In this section we discuss the problem of expanding a geometric theory $\mathbb{T}$ to a theory of presheaf type classified by the topos $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$. We shall say that a geometric theory is a *presheaf completion* of a geometric theory $\mathbb{T}$ if it is an expansion of $\mathbb{T}$ such that the geometric morphism $p_{\mathbb{T}}^{\mathbb{T}'} : \mathbf{Set}[\mathbb{T}'] \to \mathbf{Set}[\mathbb{T}]$ is isomorphic to the canonical geometric morphism

$$p_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})} : [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \to \mathbf{Set}[\mathbb{T}]$$

(cf. section 5.2.3).

Theorem 7.1.6 shows, in light of the results of section 7.1.1, that in order to obtain a presheaf completion of a given geometric theory $\mathbb{T}$ we should add a new sort $\ulcorner c \urcorner$ for each finitely presentable $\mathbb{T}$-model c which is not presented by a geometric formula over the signature of $\mathbb{T}$, and a relation symbol for each subobject of a finite product of objects of $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ which are either of the form $\mathrm{Hom}_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})}(c, -)$ (where c is not presented by any geometric formula over the signature of $\mathbb{T}$), or of the form U_A for a sort A over the signature of $\mathbb{T}$ (where U_A is the forgetful functor $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}) \to \mathbf{Set}$ at the sort A), the sort corresponding to such a representable $\mathrm{Hom}_{\mathcal{C}}(c, -)$ being $\ulcorner c \urcorner$ and to such a functor U_A being A. Indeed, by Theorem 2.1.29, the theory of the tautological structure over this extended signature will be an expansion of $\mathbb{T}$ classified by the topos $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$. Notice that if $\mathbb{T}$ satisfies the property that every finitely presentable model of $\mathbb{T}$ is presented by a geometric formula over its signature then this expansion of $\mathbb{T}$ will be localic over $\mathbb{T}$.

Of course, as is clear from the results of section 7.1.1, there are in general many ways of 'completing' a given geometric theory to a theory of presheaf type; the procedure described above represents just a particular choice which is by no means canonical. In fact, what is most interesting in practice is to obtain explicit axiomatizations of presheaf-type completions of a given theory $\mathbb{T}$ directly from the axioms of $\mathbb{T}$. Theorem 7.1.6 and the above remarks provide a useful guide in seeking such axiomatizations, as they indicate that, in order to complete a geometric theory $\mathbb{T}$ to a theory classified by the topos [f.p.$\mathbb{T}$-mod($\mathbf{Set}$), $\mathbf{Set}$], one should expand the language of $\mathbb{T}$ so as to make each finitely presentable $\mathbb{T}$-model presented by a formula over the extended signature and every property of finite tuples of elements of finitely presentable models of $\mathbb{T}$ which is preserved by $\mathbb{T}$-model homomorphisms definable by a geometric formula in the extended signature; we shall see concrete applications of this general method in Chapter 9. We note moreover that, if one is able to prove that every finitely presentable $\mathbb{T}$-model is strongly finitely presented as a model of the theory $\mathbb{T}$ when the latter is considered over a signature Σ' expanding that of $\mathbb{T}$, condition (iii) of Theorem 6.3.1 will automatically be satisfied by this latter theory, while conditions (i) and (ii) can be forced to hold by adding further axioms to $\mathbb{T}$ over the signature Σ' (cf. Theorem 8.2.10 below).

The following lemma shows that, under appropriate conditions, models which are finitely presented over a given signature remain finitely presented over a larger signature obtained from the former by adding relation symbols characterized by disjunctive sequents of a certain form.

Lemma 7.1.7 *Let $\mathbb{T}$ be a geometric theory over a signature Σ, Σ' a signature obtained from Σ by only adding relation symbols R and $\mathbb{S}$ a geometric theory over Σ' obtained from $\mathbb{T}$ by adding, for each such R, a pair of sequents of the form $(\top \vdash_{\vec{z}} R(\vec{z}) \vee \bigvee_{i\in I} \phi_i^R(\vec{z}))$ and $(R(\vec{z}) \wedge \bigvee_{i\in I} \phi_i^R(\vec{z}) \vdash_{\vec{z}} \bot)$, where, for each $i \in I$, $\phi_i^R(\vec{z})$ is a geometric formula-in-context over Σ such that there exists a geometric formula $\psi_i^R(\vec{z})$ over Σ with the property that the sequents $(\phi_i^R \wedge \psi_i^R \vdash_{\vec{z}} \bot)$ and $(\top \vdash_{\vec{z}} \phi_i^R \vee \psi_i^R)$ are provable in $\mathbb{S}$.*

Let $\mathbb{R}$ be the cartesianization of $\mathbb{S}$ (in the sense of Remark 8.2.2(c)) and $\phi(\vec{x}) = \phi(x_1, \dots, x_n)$ an $\mathbb{R}$-cartesian formula over Σ' with the property that there exists an $\mathbb{R}$-model $M_{\{\vec{x}\,.\,\phi\}}$ with n generators $\vec{\xi} = (\xi_1, \dots, \xi_n) \in [[\vec{x}\,.\,\phi]]_{M_{\{\vec{x}\,.\,\phi\}}}$ such that for any $\mathbb{R}$-model N the elements of $[[\vec{x}\,.\,\phi]]_N$ are in natural bijective correspondence with the Σ-structure homomorphisms $f : M_{\{\vec{x}\,.\,\phi\}} \to N$ such that $f(\vec{\xi}) \in [[\vec{x}\,.\,\phi]]_N$ through the assignment $f \to f(\vec{\xi})$. Then $M_{\{\vec{x}\,.\,\phi\}}$ is finitely presented by the formula $\{\vec{x}\,.\,\phi\}$ as an $\mathbb{R}$-model.

Proof We have to prove that, for any $\mathbb{R}$-model N and any tuple $\vec{a} \in [[\vec{x}\,.\,\phi]]_N$, the unique Σ-structure homomorphism $f : M_{\{\vec{x}\,.\,\phi\}} \to N$ such that $f(\vec{\xi}) = \vec{a}$ preserves the satisfaction of all the relation symbols in Σ', i.e. that for any such symbol R of arity m and any m-tuple $(y_1, \dots, y_m)$ of elements of $M_{\{\vec{x}\,.\,\phi\}}$ belonging to $M_{\{\vec{x}\,.\,\phi\}}R$, the m-tuple $(f(y_1), \dots, f(y_m))$ belongs to NR.

As $M_{\{\vec{x}\,.\,\phi\}}$ is by our hypothesis generated by the elements $\xi_1, \dots, \xi_n$, each y_i is equal to the interpretation in $M_{\{\vec{x}\,.\,\phi\}}$ of a term $t_i(\xi_1, \dots, \xi_n)$ evaluated at the tuple $\vec{\xi} = (\xi_1, \dots, \xi_n)$. For each $i \in I$, consider the formula ${\psi'}_i^R(x_1, \dots, x_n)$ obtained by making the substitution $z_i \to t_i(x_1, \dots, x_n)$ in the formula $\psi_i^R(\vec{z})$

(for each $i = 1, \ldots, m$). Let us show that the sequent $(\phi \vdash_{\vec{x}} {\psi'_i}^R)$ is valid in every $\mathbb{R}$-model. This amounts to showing that for any $\mathbb{R}$-model P and any tuple $\vec{b} \in [[\vec{x} \, . \, \phi]]_P$, $\vec{b}$ satisfies the formula ${\psi'_i}^R$. To prove this, we observe that, since $M_{\{\vec{x} \, . \, \phi\}}$ is by our hypotheses a $\mathbb{S}$-model, the m-tuple $(y_1, \ldots, y_m)$ satisfies the formula ψ_i^R (for each $i \in I$). Now, $(y_1, \ldots, y_m) \in [[\vec{y} \, . \, \psi_i^R]]_{M_{\{\vec{x} \, . \, \phi\}}}$ means that $\vec{\xi} \in [[\vec{x} \, . \, {\psi'_i}^R]]_{M_{\{\vec{x} \, . \, \phi\}}}$, which implies, since by our hypotheses there exists a Σ-structure homomorphism $M_{\{\vec{x} \, . \, \phi\}} \to P$ sending $\vec{\xi}$ to $\vec{b}$, that $\vec{b} \in [[\vec{x} \, . \, {\psi'_i}^R]]_N$, as required. Since $\mathbb{R}$ has enough set-based models as it is cartesian, we can conclude that the sequent $(\phi \vdash_{\vec{x}} {\psi'_i}^R)$ is provable in $\mathbb{R}$ and hence in $\mathbb{S}$ (for each $i \in I$). It follows that the cartesian sequent $(\phi \vdash_{\vec{x}} R(t_1/z_1, \ldots, t_m/z_m))$ is provable in $\mathbb{R}$. By evaluating this sequent in the model N at the tuple $f(\vec{\xi}) \in [[\vec{x} \, . \, \phi]]_N$ and using the fact that Σ-structure homomorphisms commute with the interpretation of Σ-terms, we obtain that $(f(y_1), \ldots, f(y_m)) = ([[\vec{x} \, . \, t_1]]_N(f(\vec{\xi})), \ldots, [[\vec{x} \, . \, t_m]]_N(f(\vec{\xi}))) \in NR$, as required. □

7.1.4 *Presheaf-type expansions*

In this section we shall consider presheaf-type expansions of presheaf-type theories and prove, under some natural hypotheses, a theorem concerning the formulae which present the models of such theories.

Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathbb{T}'$ a geometric expansion of $\mathbb{T}$ over a signature Σ'.

Suppose that $\mathcal{C}$ is a subcategory of the category of finitely presentable $\mathbb{T}'$-models such that the canonical functor

$$(p_{\mathbb{T}}^{\mathbb{T}'})_{\mathbf{Set}} : \mathbb{T}'\text{-mod}(\mathbf{Set}) \to \mathbb{T}\text{-mod}(\mathbf{Set})$$

induced by the geometric morphism $p_{\mathbb{T}}^{\mathbb{T}'} : \mathbf{Set}[\mathbb{T}'] \to \mathbf{Set}[\mathbb{T}]$ restricts to a functor

$$j : \mathcal{C} \to \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}).$$

Then we have a commutative diagram

$$\begin{tikzcd}
\mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) \arrow[r, "p_{\mathbb{T}}^{\mathbb{T}'}"] & \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \\
{[\mathcal{C}, \mathbf{Set}]} \arrow[u, "s_{\mathcal{C}}^{\mathbb{T}'}"] \arrow[r, "{[j, \mathbf{Set}]}"] & {[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}],} \arrow[u, "t^{\mathbb{T}}"']
\end{tikzcd}$$

where the geometric morphisms

$$s_{\mathcal{C}}^{\mathbb{T}'} : [\mathcal{C}, \mathbf{Set}] \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$$

and

$$t^{\mathbb{T}} : [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

are the canonical ones and $[j, \mathbf{Set}]$ is the geometric morphism canonically induced by the functor j.

Theorem 7.1.8 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ and $\mathbb{T}'$ a geometric expansion of $\mathbb{T}$ over a signature Σ'. Suppose that $\mathbb{T}'$ is classified by a topos $[\mathcal{C}, \mathbf{Set}]$, where $\mathcal{C}$ is a full subcategory of* f.p.$\mathbb{T}'$-mod($\mathbf{Set}$) *such that the functor $(p_{\mathbb{T}}^{\mathbb{T}'})_{\mathbf{Set}} : \mathbb{T}'\text{-mod}(\mathbf{Set}) \to \mathbb{T}\text{-mod}(\mathbf{Set})$ restricts to a faithful functor $j : \mathcal{C} \to$ f.p.$\mathbb{T}$-mod($\mathbf{Set}$). Then, for any model c of $\mathbb{T}'$ in $\mathcal{C}$ whose associated $\mathbb{T}$-model $j(c)$ is presented by a geometric formula $\phi(\vec{x})$ over Σ, there exists a geometric formula $\psi(\vec{x})$ over Σ' in the context $\vec{x}$ which presents the $\mathbb{T}'$-model c and such that the sequent $(\psi \vdash_{\vec{x}} \phi)$ is provable in $\mathbb{T}'$.*

Before giving the proof of the theorem, we need to recall the following well-known lemma (of which we give a proof for the reader's convenience):

Lemma 7.1.9 *Let $R : \mathcal{A} \to \mathcal{B}$ and $L : \mathcal{B} \to \mathcal{A}$ be a pair of adjoint functors, where R is the right adjoint and L is the left adjoint. Let $\eta : 1_{\mathcal{B}} \to R \circ L$ be the unit of the adjunction and b an object of $\mathcal{B}$. Then $\eta(b)$ is monic if and only if for any arrows f, g in $\mathcal{B}$ with codomain b, $Lf = Lg$ implies $f = g$. In particular, η is pointwise monic if and only if L is faithful.*

Proof Let us denote by $\tau_{a,b}$ the bijection between the sets $\mathrm{Hom}_{\mathcal{A}}(Lb, a)$ and $\mathrm{Hom}_{\mathcal{B}}(b, Ra)$ given by the adjunction. By the naturality in b of $\tau_{a,b}$, for any arrow $h : b' \to b$ in $\mathcal{B}$ the arrow $\eta_b \circ h$ corresponds under $\tau_{Lb,b'}$ to the arrow Lh. Therefore, as $\tau_{Lb,b'}$ is a bijection, $\eta_b \circ f = \eta_b \circ g$ if and only if $Lg = Lf$. From this our thesis follows at once. □

Proof (of Theorem 7.1.8) The geometric morphism

$$[j, \mathbf{Set}] : [\mathcal{C}, \mathbf{Set}] \to [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$$

is essential (cf. Example 1.1.15(d)), that is, its inverse image $[j, \mathbf{Set}]^*$ admits a left adjoint $[j, \mathbf{Set}]_!$, namely the left Kan extension along the functor j, and the following diagram (where $y_{\mathcal{C}}$ and y are the Yoneda embeddings) commutes:

$$\begin{tikzcd}
{[\mathcal{C}, \mathbf{Set}]} \arrow[r, "{[j,\mathbf{Set}]_!}"] & {[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]} \\
\mathcal{C}^{\mathrm{op}} \arrow[u, "y_{\mathcal{C}}"] \arrow[r, "j^{\mathrm{op}}"] & \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}. \arrow[u, "y"']
\end{tikzcd}$$

The functor

$$[j, \mathbf{Set}]_! : [\mathcal{C}, \mathbf{Set}] \to [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$$

satisfies the property that for any object c of $\mathcal{C}$ and any arrows $\alpha, \beta : P \to y_{\mathcal{C}}(c)$, where P is an object of $[\mathcal{C}, \mathbf{Set}]$, $[j, \mathbf{Set}]_!(\alpha) = [j, \mathbf{Set}]_!(\beta)$ implies $\alpha = \beta$. Indeed, this is clearly true for P equal to a representable by the commutativity of the above diagram, the fullness and faithfulness of the Yoneda embeddings $y_{\mathcal{C}}$ and y' and the fact that the functor j is faithful by our hypotheses, and one can always reduce to this case by considering a covering of P in $[\mathcal{C}, \mathbf{Set}]$ by representables.

Let $y_{\mathbb{T}} : \mathcal{C}_{\mathbb{T}} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ and $y_{\mathbb{T}'} : \mathcal{C}_{\mathbb{T}'} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$ be the Yoneda embeddings. By our hypotheses the geometric morphisms $s_{\mathcal{C}}^{\mathbb{T}'}$ and $t^{\mathbb{T}}$ defined above are equivalences. Therefore the geometric morphism $p_{\mathbb{T}}^{\mathbb{T}'}$ is essential and by Lemma 7.1.9 the unit of the adjunction between $(p_{\mathbb{T}}^{\mathbb{T}'})^*$ (right adjoint) and $(p_{\mathbb{T}}^{\mathbb{T}'})_!$ (left adjoint) is monic when evaluated at any object of the form $y_{\mathbb{T}'}(\{\vec{y}\,.\,\chi\})$, where $\chi(\vec{y})$ is a formula presenting a $\mathbb{T}'$-model in $\mathcal{C}$. Let c be a $\mathbb{T}'$-model in $\mathcal{C}$. Since $\mathbb{T}'$ is classified by the topos $[\mathcal{C}, \mathbf{Set}]$, c is finitely presented by a formula $\{\vec{y}\,.\,\chi\}$ over the signature Σ' of $\mathbb{T}'$. The commutativity of the above square and of the diagram preceding the statement of Theorem 7.1.8 thus implies that $(p_{\mathbb{T}}^{\mathbb{T}'})_!(y_{\mathbb{T}'}(\{\vec{y}\,.\,\chi\})) \cong y_{\mathbb{T}}(\{\vec{x}\,.\,\phi\})$, where $\phi(\vec{x})$ is a formula over the signature Σ which presents the model $j(c)$. By definition of the geometric morphism $p_{\mathbb{T}}^{\mathbb{T}'}$, we have that $(p_{\mathbb{T}}^{\mathbb{T}'})^*(y_{\mathbb{T}}(\{\vec{x}\,.\,\phi\})) = y_{\mathbb{T}'}(\{\vec{x}\,.\,\phi\})$. The unit of the adjunction between $(p_{\mathbb{T}}^{\mathbb{T}'})^*$ and $(p_{\mathbb{T}}^{\mathbb{T}'})_!$ thus yields a monic arrow

$$y_{\mathbb{T}'}(\{\vec{y}\,.\,\chi\}) \rightarrowtail {p_{\mathbb{T}}^{\mathbb{T}'}}^*({p_{\mathbb{T}}^{\mathbb{T}'}}_!(y_{\mathbb{T}'}(\{\vec{y}\,.\,\chi\}))) \cong y_{\mathbb{T}'}(\{\vec{x}\,.\,\phi\})$$

in the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$, in other words a monic arrow $\{\vec{y}\,.\,\chi\} \rightarrowtail \{\vec{x}\,.\,\phi\}$ in the geometric syntactic category $\mathcal{C}_{\mathbb{T}'}$. Therefore $\{\vec{y}\,.\,\chi\}$ is isomorphic, as an object of $\mathcal{C}_{\mathbb{T}'}$, to an object $\{\vec{x}\,.\,\phi'\}$ such that the sequent $(\phi' \vdash_{\vec{x}} \phi)$ is provable in $\mathbb{T}'$ (cf. Lemma 1.4.2); hence $\phi'(\vec{x})$ presents c as a $\mathbb{T}'$-model, as required. □

7.2 **Faithful interpretations of theories of presheaf type**

In this section we introduce the notion of faithful interpretation of geometric theories and establish sufficient criteria for theories admitting faithful interpretations of theories of presheaf type (in the sense of the definition below) to be again of presheaf type.

7.2.1 *General results*

Definition 7.2.1 A geometric morphism $a : \mathbf{Set}[\mathbb{T}] \to \mathbf{Set}[\mathbb{T}']$ between the classifying toposes of two geometric theories $\mathbb{T}'$ and $\mathbb{T}$ is said to be a *faithful interpretation* of $\mathbb{T}'$ into $\mathbb{T}$ if for every Grothendieck topos $\mathcal{E}$ the induced functor

$$a_{\mathcal{E}} : \mathbb{T}\text{-mod}(\mathcal{E}) \to \mathbb{T}'\text{-mod}(\mathcal{E})$$

between the categories of models is faithful and full on isomorphisms.

For any faithful interpretation of $\mathbb{T}'$ into $\mathbb{T}$ and any Grothendieck topos $\mathcal{E}$, we have a subcategory $\mathrm{Im}(a_{\mathcal{E}})$ of $\mathbb{T}'$-mod$(\mathcal{E})$ whose objects (resp. arrows) are exactly the models (resp. the model homomorphisms) in the image of the functor $a_{\mathcal{E}}$. Notice that for any functor $F : \mathcal{A} \to \mathbb{T}\text{-mod}(\mathcal{E})$, F is full and faithful if and only if its composite $a_{\mathcal{E}} \circ F$ with $a_{\mathcal{E}}$ is full and faithful as a functor $\mathcal{A} \to \mathrm{Im}(a_{\mathcal{E}})$.

Remarks 7.2.2 (a) Any quotient $\mathbb{S}$ of a geometric theory $\mathbb{T}$ defines a faithful interpretation of $\mathbb{T}$ into $\mathbb{S}$.

(b) The *injectivization* $\mathbb{T}_m$ (in the sense of Definition 7.2.10) of a geometric theory $\mathbb{T}$ defines a faithful interpretation of $\mathbb{T}$ into $\mathbb{T}_m$.

Given a faithful interpretation of $\mathbb{T}'$ into $\mathbb{T}$ as above, we can reformulate the conditions of Theorem 6.3.1 for the theory $\mathbb{T}$ with respect to a full subcategory $\mathcal{K}$ of f.p.$\mathbb{T}$-mod(**Set**) in alternative ways, as follows.

Condition (iii)(c) of Theorem 6.3.1 for $\mathbb{T}$ with respect to $\mathcal{K}$ can be reformulated as follows: the composite functor

$$q_{\mathcal{E}} := a_{\mathcal{E}} \circ u_{\mathcal{K}} : \mathbf{Flat}(\mathcal{K}^{\text{op}}, \mathcal{E}) \to \mathbb{T}'\text{-mod}(\mathcal{E})$$

is full and faithful into the image $\text{Im}(a_{\mathcal{E}})$ of $a_{\mathcal{E}}$, where $u_{\mathcal{K}}$ is the functor

$$\mathbf{Flat}(\mathcal{K}^{\text{op}}, \mathcal{E}) \to \mathbb{T}\text{-mod}(\mathcal{E})$$

induced by the canonical geometric morphism $p_{\mathcal{K}} : [\mathcal{K}, \mathbf{Set}] \to \mathbf{Set}[\mathbb{T}]$.

Condition (ii) of Theorem 6.3.1 for $\mathbb{T}$ with respect to $\mathcal{K}$ can be reformulated by saying that for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, the canonical morphism

$$q_{\mathcal{E}}(H_M) \to a_{\mathcal{E}}(M)$$

is an isomorphism. Indeed, this morphism is the image under $a_{\mathcal{E}}$ of the canonical morphism $u_{\mathcal{K}}(H_M) \to M$ considered in Remark 6.3.2(c).

Suppose moreover that the functor

$$a_{\mathbf{Set}} : \mathbb{T}\text{-mod}(\mathbf{Set}) \to \mathbb{T}'\text{-mod}(\mathbf{Set})$$

restricts to a functor

$$f : \mathcal{K} \to \mathcal{H},$$

where $\mathcal{H}$ is a full subcategory of f.p.$\mathbb{T}'$-mod(**Set**). We have a commutative diagram

$$\begin{tikzcd} {[\mathcal{K}, \mathbf{Set}]} \arrow[r, "{[f, \mathbf{Set}]}"] \arrow[d, "p_{\mathcal{K}}"] & {[\mathcal{H}, \mathbf{Set}]} \arrow[d, "p_{\mathcal{H}}"] \\ \mathbf{Set}[\mathbb{T}] \arrow[r, "a"] & \mathbf{Set}[\mathbb{T}'], \end{tikzcd}$$

where $p_{\mathcal{K}}$ (resp. $p_{\mathcal{H}}$) is the canonical geometric morphism determined by the universal property of the classifying topos of $\mathbb{T}$ (resp. of $\mathbb{T}'$).

Also, for any $\mathbb{T}'$-model M in a Grothendieck topos $\mathcal{E}$, the functors $a_{\mathcal{E}/E}$ (for $E \in \mathcal{E}$) induce an $\mathcal{E}$-indexed functor

$$\int a : \underline{\int H_M^{\mathbb{T}}}_{\mathcal{E}} \to \underline{\int H_{a_{\mathcal{E}}(M)}^{\mathbb{T}'}}_{\mathcal{E}}$$

between the $\mathcal{E}$-indexed categories of elements $\underline{\int H_M^{\mathbb{T}}}_{\mathcal{E}}$ and $\underline{\int H_{a_{\mathcal{E}}(M)}^{\mathbb{T}'}}_{\mathcal{E}}$ of the functors

$$H^{\mathbb{T}}_{M} := \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^{*}_{\mathcal{E}}(-), M) : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$$

and

$$H^{\mathbb{T}'}_{a_{\mathcal{E}}(M)} := \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}'\text{-mod}(\mathcal{E})}}(\gamma^{*}_{\mathcal{E}}(-), a_{\mathcal{E}}(M)) : \mathcal{H}^{\mathrm{op}} \to \mathcal{E}.$$

Since a is by our hypothesis a faithful interpretation of theories, $\int a$ is faithful at each fibre. Suppose now that $\int a$ is moreover $\mathcal{E}$-full (in the sense of Definition 5.1.8). Since for any functor F with values in a Grothendieck topos, F is flat if and only if its $\mathcal{E}$-indexed category of elements is $\mathcal{E}$-filtered, we conclude by Proposition 5.1.10 that if $\int a$ is $\mathcal{E}$-final then the theory $\mathbb{T}$ satisfies condition (i) of Theorem 6.3.1 with respect to the category $\mathcal{K}$ if $\mathbb{T}'$ does so with respect to the category $\mathcal{H}$. Moreover, Proposition 5.2.14 and Theorem 5.1.11 ensure that if $\mathbb{T}'$ satisfies condition (ii) of Theorem 6.3.1 with respect to the category $\mathcal{H}$ then $\mathbb{T}$ does so with respect to the category $\mathcal{K}$.

Summarizing, we have the following result:

Theorem 7.2.3 *Let $a : \mathbf{Set}[\mathbb{T}] \to \mathbf{Set}[\mathbb{T}']$ be a faithful interpretation of geometric theories and let $\mathcal{K}$ and $\mathcal{H}$ be full subcategories respectively of* f.p.$\mathbb{T}$-mod($\mathbf{Set}$) *and of* f.p.$\mathbb{T}'$-mod($\mathbf{Set}$) *such that the functor*

$$a_{\mathbf{Set}} : \mathbb{T}\text{-mod}(\mathbf{Set}) \to \mathbb{T}'\text{-mod}(\mathbf{Set})$$

restricts to a functor $\mathcal{K} \to \mathcal{H}$ and for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, the $\mathcal{E}$-indexed functor

$$\int a : \underline{\int H^{\mathbb{T}}_{M}}_{\mathcal{E}} \to \underline{\int H^{\mathbb{T}'}_{a_{\mathcal{E}}(M)}}_{\mathcal{E}}$$

is $\mathcal{E}$-full and $\mathcal{E}$-final.

Then the theory $\mathbb{T}$ satisfies condition (i) *(resp. condition (ii)) of Theorem 6.3.1 with respect to the category $\mathcal{K}$ if $\mathbb{T}'$ satisfies condition (i) (resp. condition (ii)) of Theorem 6.3.1 with respect to the category $\mathcal{H}$.* □

Remark 7.2.4 If $\mathbb{T}$ is the injectivization of $\mathbb{T}'$ (in the sense of Definition 7.2.10) or if $\mathbb{T}$ is a quotient of $\mathbb{T}'$ then the $\mathcal{E}$-indexed functor $\int a$ in the statement of Theorem 7.2.3 is $\mathcal{E}$-full. Indeed, in the first case this follows from the fact that if the composite of two arrows is monic then the first one is monic, while in the second it follows from the fact that the category of models of $\mathbb{T}$ in any Grothendieck topos is a full subcategory of the category of models of $\mathbb{T}'$ in it.

The following proposition provides an explicit rephrasing of the finality condition for the $\mathcal{E}$-indexed functor $\int a$ of Theorem 7.2.3.

Proposition 7.2.5 *Let $a : \mathbf{Set}[\mathbb{T}] \to \mathbf{Set}[\mathbb{T}']$ be a faithful interpretation of geometric theories and let M a $\mathbb{T}$-model in a Grothendieck topos $\mathcal{E}$ such that the functor $H^{\mathbb{T}'}_{a_{\mathcal{E}}(M)}$ is flat. Let Σ' be the signature of $\mathbb{T}'$. Then the following conditions are equivalent:*

(i) *The functor* $\int a$ *is* $\mathcal{E}$*-final.*

(ii) *For any object* E *of* $\mathcal{E}$*,* $\mathbb{T}'$*-model* c *in* $\mathcal{H}$*,* $\mathbb{T}$*-model* M *in* $\mathcal{E}$ *and* $\mathbb{T}'$*-model homomorphism* $x : \gamma^*_{\mathcal{E}/E}(c) \to !^*_E(a_{\mathcal{E}}(M))$*, there exist an epimorphic family* $\{e_i : E_i \to E \mid i \in I\}$ *in* $\mathcal{E}$ *and for each* $i \in I$ *a* $\mathbb{T}'$*-model homomorphism* $f_i : c \to a_{\mathbf{Set}}(c_i)$*, where* c_i *is a* $\mathbb{T}$*-model in* $\mathcal{K}$*, and a* $\mathbb{T}'$*-model homomorphism* $x_i : \gamma^*_{\mathcal{E}/E_i}(a_{\mathbf{Set}}(c_i)) \to !_{E_i}(a_{\mathcal{E}}(M))$ *such that* $x_i \circ \gamma^*_{\mathcal{E}/E_i}(f_i) = e_i^*(x)$*.*

(iii) *For any object* E *of* $\mathcal{E}$ *and any* Σ'*-structure homomorphism* $x : c \to \mathrm{Hom}_{\mathcal{E}}(E, a_{\mathcal{E}}(M))$*, where* c *is a* $\mathbb{T}'$*-model in* $\mathcal{H}$*, there exist an epimorphic family* $\{e_i : E_i \to E \mid i \in I\}$ *in* $\mathcal{E}$ *and for each* $i \in I$ *a* $\mathbb{T}'$*-model homomorphism* $f_i : c \to a_{\mathbf{Set}}(c_i)$*, where* c_i *is a* $\mathbb{T}$*-model in* $\mathcal{K}$*, and a* Σ'*-structure homomorphism* $x_i : a_{\mathbf{Set}}(c_i) \to \mathrm{Hom}_{\mathcal{E}}(E_i, a_{\mathcal{E}}(M))$ *such that* $x_i \circ f_i = \mathrm{Hom}_{\mathcal{E}}(e_i, a_{\mathcal{E}}(M)) \circ x$*.*

Proof The equivalence between the first two conditions immediately follows from the fact that, H_M being flat, the $\mathcal{E}$-indexed category $\underline{\int H_M}_{\mathcal{E}}$ is $\mathcal{E}$-filtered and hence Remark 5.1.9(c) applies, while the equivalence between the second and the third conditions follows from Proposition 6.2.3. □

Remark 7.2.6 If $\mathcal{E} = \mathbf{Set}$ and $\mathbb{T}$ is a quotient of $\mathbb{T}'$ then condition (iii) of Proposition 7.2.5 rewrites as follows: for any $\mathbb{T}$-model M in $\mathbf{Set}$, any $\mathbb{T}'$-model c in $\mathcal{H}$ and any $\mathbb{T}'$-model homomorphism $f : c \to M$, there exist a $\mathbb{T}$-model d in $\mathcal{K}$ and $\mathbb{T}'$-model homomorphisms $g : c \to d$ and $h : d \to M$ such that $h \circ g = f$. In the particular case where $\mathbb{T}'$ is the empty theory over a signature Σ, $\mathbb{T}$ is an $\mathcal{F}$-finitary geometric theory over Σ (in the sense of Definition 3.1 [73]), $\mathcal{H}$ is the category of finite Σ-structures and $\mathcal{K}$ is the category of finite $\mathbb{T}$-models, the condition, required for every $\mathbb{T}$-model M in $\mathbf{Set}$, specializes precisely to the 'finite structure condition' of [73].

The following result shows a relation between the action on points of a morphism of classifying toposes and the related induced action on flat functors:

Proposition 7.2.7 *Let* $a : \mathbf{Set}[\mathbb{T}] \to \mathbf{Set}[\mathbb{T}']$ *be a faithful interpretation of geometric theories and* $\mathcal{K}$ *and* $\mathcal{H}$ *full subcategories respectively of* f.p.$\mathbb{T}$-mod($\mathbf{Set}$) *and of* f.p.$\mathbb{T}'$-mod($\mathbf{Set}$) *such that the functor*

$$a_{\mathbf{Set}} : \mathbb{T}\text{-mod}(\mathbf{Set}) \to \mathbb{T}'\text{-mod}(\mathbf{Set})$$

restricts to a functor $f : \mathcal{K} \to \mathcal{H}$*. Then, for every Grothendieck topos* $\mathcal{E}$*, the extension functor*

$$\mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E}) \to \mathbf{Flat}(\mathcal{H}^{\mathrm{op}}, \mathcal{E}) \to \mathbb{T}'\text{-mod}(\mathcal{E})$$

along the geometric morphism $[f, \mathbf{Set}] : [\mathcal{K}, \mathbf{Set}] \to [\mathcal{H}, \mathbf{Set}]$ *(as in section 5.2.2) takes values in the subcategory* $\mathrm{Im}(a_{\mathcal{E}})$ *of* $\mathbb{T}'$-mod($\mathcal{E}$)*.*

Proof The diagram

$$\begin{tikzcd}
{[\mathcal{H}, \mathbf{Set}]} \arrow[r, "p_{\mathcal{H}}"] & \mathbf{Set}[\mathbb{T}'] \\
{[\mathcal{K}, \mathbf{Set}]} \arrow[r, "p_{\mathcal{K}}"] \arrow[u, "{[f, \mathbf{Set}]}"] & \mathbf{Set}[\mathbb{T}] \arrow[u, "a"']
\end{tikzcd}$$

clearly commutes by definition of the functor f. Therefore, by Diaconescu's equivalence, for any Grothendieck topos $\mathcal{E}$ we have a commutative diagram

$$\begin{tikzcd}
\mathbf{Flat}(\mathcal{H}^{\mathrm{op}}, \mathcal{E}) \arrow[r, "u^{\mathbb{T}'}_{(\mathcal{H},\mathcal{E})}"] & \mathbb{T}'\text{-mod}(\mathcal{E}) \\
\mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E}) \arrow[r, "u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})}"] \arrow[u, "\xi_{\mathcal{E}}"] & \mathbb{T}\text{-mod}(\mathcal{E}), \arrow[u, "a_{\mathcal{E}}"']
\end{tikzcd}$$

where $u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})}$ and $u^{\mathbb{T}'}_{(\mathcal{H},\mathcal{E})}$ are the functors of section 5.2.3 and $\xi_{\mathcal{E}}$ is the extension functor induced by the geometric morphism $[f, \mathbf{Set}]$ as in section 5.2.2. From this our thesis immediately follows. □

The following result is a particular case of Proposition 7.2.7, obtained by taking $\mathbb{T}'$ to be the geometric theory of flat functors on $\mathcal{C}^{\mathrm{op}}$, $\mathbb{T}$ its injectivization (in the sense of Definition 7.2.10), $\mathcal{H}$ equal to $\mathcal{C}$ and $\mathcal{K}$ equal to $\mathcal{D}$.

Recall that an object a in a topos $\mathcal{E}$ is said to be *decidable* if the diagonal subobject $a \rightarrowtail a \times a$ has a complement in $\mathrm{Sub}_{\mathcal{E}}(a \times a)$.

Corollary 7.2.8 *Let $\mathcal{D}$ be a subcategory of a small category $\mathcal{C}$ and $\mathcal{E}$ a Grothendieck topos. If all the arrows in $\mathcal{D}$ are monic in $\mathcal{C}$ then*

(i) *For any flat functor $F : \mathcal{D}^{\mathrm{op}} \to \mathcal{E}$ and any object $d \in \mathcal{D}$, $\tilde{F}(d)$ is a decidable object of $\mathcal{E}$.*

(ii) *For any natural transformation $\alpha : F \to G$ between two flat functors $F, G : \mathcal{D}^{\mathrm{op}} \to \mathcal{E}$, $\tilde{\alpha}(c) : \tilde{F}(c) \to \tilde{G}(c)$ is monic in $\mathcal{E}$ for every $c \in \mathcal{C}$; in particular, $\alpha(d) : F(d) \to G(d)$ is monic in $\mathcal{E}$ for every $d \in \mathcal{D}$.*

□

The following corollary is obtained by applying Proposition 7.2.7 in the case when $\mathbb{T}$ is equal to the injectivization of $\mathbb{T}'$ (in the sense of Definition 7.2.10), $\mathcal{H}$ is a full subcategory of f.p.$\mathbb{T}'$-mod($\mathbf{Set}$) and $\mathcal{K} = \mathcal{H}$.

Corollary 7.2.9 *Let $\mathbb{T}$ be a theory of presheaf type classified by a topos $[\mathcal{K}, \mathbf{Set}]$, where $\mathcal{K}$ is a full subcategory of* f.p.$\mathbb{T}$-mod($\mathbf{Set}$)*. If all the $\mathbb{T}$-model homomorphisms in $\mathcal{K}$ are sortwise injective then every $\mathbb{T}$-model homomorphism in a Grothendieck topos is sortwise monic.* □

7.2.2 *Injectivizations of theories*

For any geometric theory $\mathbb{T}$, we can slightly modify its syntax so as to obtain a geometric theory whose models in **Set** are the same as those of $\mathbb{T}$ and whose model homomorphisms are precisely the sortwise injective $\mathbb{T}$-model homomorphisms. This construction is useful in many contexts. For instance, in section 8.1.1 we show that if the category f.p.$\mathbb{T}$-mod(**Set**) of finitely presentable models of a theory of presheaf type $\mathbb{T}$ satisfies the amalgamation property (i.e. the dual of the right Ore condition, cf. Definition 10.4.1) then the quotient of $\mathbb{T}$ corresponding to the subtopos $\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}})$ of $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ (where J_{at} is the atomic topology) via the duality of Theorem 3.2.5 axiomatizes the homogeneous $\mathbb{T}$-models (in the sense of section 8.1.1) in any Grothendieck topos. Now, the notion of homogeneous $\mathbb{T}$-model, which is strictly related to the notion of weakly homogeneous model in classical Model Theory (cf. [48]), is mostly interesting when the arrows of the category $\mathbb{T}$-mod(**Set**) are all monic; indeed, as shown in section 10.5, a necessary condition for $\mathbb{T}$ to admit an associated 'concrete' Galois theory (i.e. a Galois equivalence holding at the level of sites) is that all the arrows in f.p.$\mathbb{T}$-mod(**Set**) be strict monomorphisms.

This motivates the following definition:

Definition 7.2.10 Let $\mathbb{T}$ be a geometric theory over a signature Σ. The *injectivization* $\mathbb{T}_m$ of $\mathbb{T}$ is the geometric theory obtained from $\mathbb{T}$ by adding a binary predicate $D_A \rightarrowtail A, A$ for each sort A of Σ and the coherent sequents

$$(D_A(x^A, y^A) \wedge x^A = y^A) \vdash_{x^A, y^A} \bot)$$

and

$$(\top \vdash_{x^A, y^A} D_A(x^A, y^A) \vee x^A = y^A).$$

The models of $\mathbb{T}_m$ in an arbitrary topos $\mathcal{E}$ coincide with the models M of $\mathbb{T}$ in $\mathcal{E}$ which are sortwise decidable, in the sense that for every sort A of the signature of $\mathbb{T}$ the object MA of $\mathcal{E}$ is *decidable*, i.e. the diagonal subobject of MA is complemented (by the interpretation of D_A in M).

As shown by the following lemma, the arrows $M \to N$ in the category $\mathbb{T}_m$-mod(**Set**) are precisely the $\mathbb{T}$-model homomorphisms $f : M \to N$ such that for every sort A over the signature of $\mathbb{T}$, the arrow $f_A : MA \to NA$ is a monomorphism in $\mathcal{E}$:

Lemma 7.2.11 *Let A and B be decidable objects in a topos $\mathcal{E}$ and $f : A \to B$ an arrow in $\mathcal{E}$. Let $D_A \rightarrowtail A \times A$ and $D_B \rightarrowtail B \times B$ be respectively the complements of the diagonal subobjects $\delta_A : A \rightarrowtail A \times A$ and $\delta_B : B \rightarrowtail B \times B$. Then f is a monomorphism if and only if the arrow $f \times f : A \times A \to B \times B$ restricts to an arrow $D_A \to D_B$.*

Proof It is immediate to see that $f : A \to B$ is a monomorphism if and only if the diagram

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\delta_A"] & B \arrow[d, "\delta_B"] \\ A \times A \arrow[r, "f \times f"] & B \times B \end{tikzcd}$$

is a pullback square.

Since pullback functors preserve arbitrary unions and intersections of subobjects in a topos (as they have both a left and a right adjoint), we have that $(f \times f)^*(D_B) \cong \neg(f \times f)^*(\delta_B)$. Now, $f \times f : A \times A \to B \times B$ restricts to an arrow $D_A \to D_B$ if and only if $D_A \leq (f \times f)^*(D_B)$. But this condition holds if and only if $D_A \cap (f \times f)^*(\delta_B) \cong 0$, i.e. if and only if $(f \times f)^*(\delta_B) \leq \delta_A$, which is equivalent to the condition $(f \times f)^*(\delta_B) \cong \delta_A$ (as $\delta_A \leq (f \times f)^*(\delta_B)$). □

Several injectivizations of theories of presheaf type have been considered in the literature, e.g. in [63] and in [55]; see also sections 10.4 and 10.5 for other examples arising in the context of topos-theoretic Galois-type equivalences.

As we shall see in section 7.2.5, under certain conditions, the injectivization of a theory of presheaf type is again of presheaf type.

Lemma 7.2.12 *Let $\{f_i : A_i \to B \mid i \in I\}$ be a family of arrows in a Grothendieck topos $\mathcal{E}$. Then the arrow $\coprod_{i\in I} f_i : \coprod_{i\in I} A_i \to B$ is monic if and only if, for every $i \in I$, f_i is monic and, for every $i, i' \in I$, either $i = i'$ or the subobjects $f_i : A_i \rightarrowtail B$ and $f_i' : A_{i'} \rightarrowtail B$ are disjoint.*

Proof The 'if' direction follows from Proposition IV 7.6 [63], while the 'only if' one follows from the fact that coproducts in a topos are always disjoint (cf. for instance Corollary IV 10.5 [63]). □

Remark 7.2.13 If in the statement of Lemma 7.2.12 the objects A_i are all equal to the terminal object $1_{\mathcal{E}}$ of $\mathcal{E}$ then the arrows f_i are automatically monic and any two of them are disjoint if and only if their equalizer is zero.

The following proposition is a corollary of Theorem 6.2.1:

Proposition 7.2.14 *Let $\mathbb{T}$ be a geometric theory. Then, for any $\mathbb{T}$-models M and N in a Grothendieck topos $\mathcal{E}$ which are sortwise decidable, there exists an object $\underline{\mathrm{Hom}^{\mathcal{E}}_{\mathbb{T}_m\text{-mod}(\mathcal{E})}}(M,N)$ of $\mathcal{E}$ satisfying the following universal property: for any object E of $\mathcal{E}$ we have an equivalence*

$$\mathrm{Hom}_{\mathcal{E}}(E, \underline{\mathrm{Hom}^{\mathcal{E}}_{\mathbb{T}_m\text{-mod}(\mathcal{E})}}(M,N)) \cong \mathrm{Hom}_{\mathbb{T}_m\text{-mod}(\mathcal{E}/E)}(!_E^*(M), !_E^*(N))$$

natural in $E \in \mathcal{E}$.

Remarks 7.2.15 (a) For any $\mathcal{E}$, M and N, $\underline{\mathrm{Hom}^{\mathcal{E}}_{\mathbb{T}_m\text{-mod}(\mathcal{E})}}(M,N)$ embeds canonically as a subobject of $\underline{\mathrm{Hom}^{\mathcal{E}}_{\mathbb{T}\text{-mod}(\mathcal{E})}}(M,N)$.

(b) Let $\mathbb{T}$ be a geometric theory over a signature Σ, c a finitely presentable $\mathbb{T}$-model and M a sortwise decidable model of $\mathbb{T}$ in a Grothendieck topos $\mathcal{E}$. Then the subobject

$$\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}_m\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M) \rightarrowtail \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M)$$

can be identified with the interpretation of the formula

$$\chi_c(f) := \bigwedge_{\substack{A \text{ sort of } \Sigma, \\ x,y \in cA,\ x \neq y}} (\pi_A(f)(\gamma^*_{\mathcal{E}}(\overline{x})), \pi_A(f)(\gamma^*_{\mathcal{E}}(\overline{y}))) \in D_{MA},$$

written in the internal language of the topos $\mathcal{E}$, where

$$\pi_A : \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N) \to NA^{MA}$$

is the arrow defined in Remark 6.2.2(b) and $\overline{x}, \overline{y} : 1 \to cA$ are the arrows in **Set** corresponding respectively to the elements x and y of cA. Indeed, a $\mathbb{T}$-model homomorphism $f : \gamma^*_{\mathcal{E}}(c) \to M$ is sortwise monic if and only if, for every sort A of Σ and any distinct elements $x, y \in cA$, $f_A(\gamma^*_{\mathcal{E}}(\overline{x}))$ and $f_A(\gamma^*_{\mathcal{E}}(\overline{y}))$ are disjoint, i.e. they satisfy the relation D_{MA}. It follows that if for every finitely presentable $\mathbb{T}$-model c, the formula $\chi_c(f)$ is equivalent to a geometric formula over the signature of $\mathbb{T}_m$ (e.g. when Σ only contains a finite number of sorts and for every sort A and finitely presentable $\mathbb{T}$-model c the set cA is finite), then the theory $\mathbb{T}_m$ satisfies condition (iii)(a) of Theorem 6.3.1 with respect to every finitely presentable $\mathbb{T}$-model if $\mathbb{T}$ does.

Lemma 7.2.16 *Let $\mathbb{T}$ be a geometric theory over a signature Σ, c a set-based $\mathbb{T}$-model and M a $\mathbb{T}$-model in a Grothendieck topos $\mathcal{E}$. Then any generalized element $x : E \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}_m\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M)$ can be identified with a Σ-structure homomorphism $\xi_x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ which is* sortwise disjunctive *in the sense that, for every sort A of Σ, the function $(\xi_x)_A : cA \to \mathrm{Hom}_{\mathcal{E}}(E, MA)$ has the property that for any distinct elements $z, w \in cA$, the arrows $(\xi_x)_A(z), (\xi_x)_A(w) : E \to MA$ have equalizer zero in $\mathcal{E}$.*

Proof The thesis follows from Proposition 6.2.3 by observing that, by Lemma 7.2.12 and Remark 7.2.13, an arrow $\tau_A : \gamma^*_{\mathcal{E}/E}(cA) \to !^*_E(MA)$ is monic in $\mathcal{E}/E$ if and only if the corresponding function $\xi_A : cA \to \mathrm{Hom}_{\mathcal{E}}(E, MA)$ satisfies the property that for any distinct elements $z, w \in cA$, the arrows $\xi_A(z) : E \to MA$ and $\xi_A(w) : E \to MA$ have equalizer zero in $\mathcal{E}$ (notice that two arrows $s, t : (a : A \to E) \to (b : B \to E)$ in $\mathcal{E}/E$ have equalizer zero in $\mathcal{E}/E$ if and only if the arrows $s : A \to B$ and $t : A \to B$ have equalizer zero in $\mathcal{E}$). □

7.2.3 *Finitely presentable and finitely generated models*

In this section we discuss, for the purpose of establishing our main criteria for the injectivization of a theory of presheaf type to be again of presheaf type, the relationship between finitely presentable and finitely generated models of a given geometric theory.

Throughout this section, we shall assume for simplicity all our first-order signatures to be one-sorted, if not otherwise stated; it is certainly possible to extend our definitions and results to the general situation, but we shall not embark on the routine task of making this explicit.

Let Σ be a (one-sorted) first-order signature. Recall that, given a Σ-structure M and a subset $A \subseteq M$, the Σ-structure $< A >$ generated by A is the smallest Σ-substructure of M containing A; it can be concretely represented as the union $\bigcup\limits_{n\in\mathbb{N}} A_n$, where the subsets A_n are defined inductively as follows:

$$A_0 = A \cup \{Mc \mid c \text{ constant of } \Sigma\},$$

and

$$A_{n+1} = A_n \cup \{(Mf)(\vec{a}) \mid f \text{ is an } m\text{-ary function symbol of } \Sigma \text{ for some } m > 0 \text{ and } \vec{a} \text{ is an } m\text{-tuple of elements of } A_n\}.$$

A Σ-structure M is said to be *finitely generated* if there exists a finite subset $A \subseteq M$ such that $M =< A >$.

We shall say that a set-based model N of a geometric theory $\mathbb{T}$ is a *quotient* of a set-based $\mathbb{T}$-model M if there exists a surjective $\mathbb{T}$-model homomorphism $M \to N$.

Proposition 7.2.17 *Let $\mathbb{T}$ be a geometric theory over a signature Σ. Then every quotient in* $\mathbb{T}$-mod(**Set**) *of a finitely generated $\mathbb{T}$-model is finitely generated.*

Proof Let $q : M \to N$ be a surjective homomorphism of $\mathbb{T}$-models. Suppose that M is finitely generated, by a finite subset $A \subseteq M$. Set $B = q(A)$. This set is clearly finite, and N is equal to $< B >$. □

Proposition 7.2.18 *Let $\mathbb{T}$ be a geometric theory over a signature Σ whose axioms are all of the form $(\phi \vdash_{\vec{x}} \psi)$, where ψ is a quantifier-free geometric formula. Then every finitely presentable $\mathbb{T}$-model is finitely generated.*

Proof First, we notice that any geometric theory $\mathbb{T}$ over a signature Σ such that all its axioms are of the form $(\phi \vdash_{\vec{x}} \psi)$, where ψ is a quantifier-free geometric formula, satisfies the property that every substructure of a model of $\mathbb{T}$ is a model of $\mathbb{T}$.

Let M be a finitely presentable $\mathbb{T}$-model. We can clearly represent M as the directed union of its finitely generated substructures (equivalently, submodels). Since M is finitely presentable, the identity on M factors through one of the embeddings of a finitely generated $\mathbb{T}$-submodel of M into M; such an embedding is thus necessarily an isomorphism, whence M is finitely generated, as required. □

Recall that a category $\mathcal{A}$ is said to be *finitely accessible* if it has filtered colimits and a set P of finitely presentable objects such that every object of $\mathcal{A}$ is a filtered colimit of objects in P, equivalently if it is of the form Ind-$\mathcal{C}$ for some small category $\mathcal{C}$.

Proposition 7.2.19 *Let $\mathbb{T}$ be a geometric theory over a signature Σ whose category of set-based models is finitely accessible (for instance, a theory of presheaf type). Then every finitely generated $\mathbb{T}$-model is a quotient of a finitely presentable $\mathbb{T}$-model.*

If moreover all the axioms of $\mathbb{T}$ are of the form $(\phi \vdash_{\vec{x}} \psi)$, where ψ is a quantifier-free geometric formula, then the finitely generated $\mathbb{T}$-models are precisely the quotients of the finitely presentable $\mathbb{T}$-models in the category $\mathbb{T}$-mod(**Set**).

Proof As the category $\mathbb{T}$-mod(**Set**) is finitely accessible, M can be expressed as a filtered colimit $(M = \mathrm{colim}(N_i), \{J_i : N_i \to M \mid i \in \mathcal{I}\})$ of finitely presentable $\mathbb{T}$-models N_i. Let $a_1, \ldots, a_n$ be a finite set of generators for M. As the family of arrows $\{J_i : N_i \to M \mid i \in \mathcal{I}\}$ is jointly surjective (since filtered colimits in $\mathbb{T}$-mod(**Set**) are computed sortwise as in **Set**), for any generator a_j there exist an index $k_j \in \mathcal{I}$ and an element d_j of N_{k_j} such that $J_{k_j}(d_j) = a_j$. Now, the set of generators a_j being finite and the indexing category $\mathcal{I}$ being filtered, we can suppose without loss of generality that all the k_j are equal. We thus have an index $k \in \mathcal{I}$ and a string of elements of N_k which are sent by J_k to the generators of M, whence the homomorphism $J_k : N_k \to M$ is surjective.

The second part of the proposition follows from Propositions 7.2.17 and 7.2.18, which ensure that every quotient of a finitely presentable $\mathbb{T}$-model is finitely generated. $\square$

Proposition 7.2.20 *Let Σ be a signature with no relation symbols and $\mathbb{T}$ a geometric theory over a larger signature Σ' obtained from Σ by only adding relation symbols whose interpretation in any set-based $\mathbb{T}$-model coincides with the complement of that of a geometric formula over Σ (e.g. $\mathbb{T}$ can be the injectivization of a geometric theory over Σ). Suppose that all the $\mathbb{T}$-model homomorphisms between set-based $\mathbb{T}$-models are injective. Then every finitely generated $\mathbb{T}$-model is finitely presentable.*

Proof We observed in Lemma 6.1.12 that, given a category $\mathcal{D}$ with filtered colimits, an object M of $\mathcal{D}$ is finitely presentable in $\mathcal{D}$ if and only if, for any diagram $D : \mathcal{I} \to \mathcal{D}$ defined on a small filtered category $\mathcal{I}$, every arrow $M \to \mathrm{colim}(D)$ factors through one of the canonical colimit arrows $J_i : D(i) \to \mathrm{colim}(D)$ and, for any arrows $f : M \to D(i)$ and $g : M \to D(j)$ in $\mathcal{D}$ such that $J_i \circ f = J_j \circ g$, there exist $k \in \mathcal{I}$ and two arrows $s : i \to k$ and $t : j \to k$ in $\mathcal{I}$ such that $D(s) \circ f = D(t) \circ g$. Now, the latter condition is automatically satisfied if all the arrows of $\mathcal{D}$ are monic (since the category $\mathcal{I}$ is filtered); in particular, applying this criterion in the case of our theory $\mathbb{T}$ yields the following characterization of its finitely presentable models: a set-based $\mathbb{T}$-model M is finitely presentable if and only if for any filtered colimit $\mathrm{colim}(N_i)$ of set-based $\mathbb{T}$-models with colimit arrows $J_i : N_i \to \mathrm{colim}(N_i)$, every $\mathbb{T}$-model homomorphism $f : M \to \mathrm{colim}(N_i)$ factors through at least one arrow J_i in the category $\mathbb{T}$-mod(**Set**).

Now, if M is finitely generated by elements $a_1, \ldots, a_n$, for any $i \in \{1, \ldots, n\}$ there exist an object k_i of $\mathcal{I}$ and an element $b_i \in N_{k_i}$ such that $f(a_i) = J_i(b_i)$. As the category $\mathcal{I}$ is filtered, we can suppose without loss of generality that all the k_i are equal to each other. Therefore there exist an object $k \in \mathcal{I}$ and elements $b_1, \ldots, b_n$ of N_k such that $f(a_i) = J_k(b_i)$ for all i. The image of M under f is thus entirely contained in N_k and hence we have a function $g : M \to N_k$ such that $J_k \circ g = f$. Let us prove that g is a Σ'-structure homomorphism. The fact that g is a Σ-structure homomorphism, i.e. that it commutes with the interpretation of the function symbols of Σ, follows from the fact that f is so, by virtue of the injectivity of J_k. The fact that g preserves the satisfaction of atomic relations over Σ' follows from the fact that f is a Σ'-structure homomorphism, since the interpretation of any of these relations in a set-based $\mathbb{T}$-model is the complement of that of a geometric formula over Σ. $\square$

A useful sufficient condition for condition (iii)(b)-(1) of Remark 6.3.2(f) to hold, which is applicable to categories $\mathcal{K}$ whose arrows are not necessarily monomorphisms, is given by the following result:

Theorem 7.2.21 *Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathcal{K}$ a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**) *such that every model in $\mathcal{K}$ is a quotient of a $\mathbb{T}$-model which is finitely presented by a geometric formula over Σ. Then $\mathbb{T}$ satisfies condition (iii)(b)-(1) of Remark 6.3.2(f) with respect to the category $\mathcal{K}$.*

Proof We have to show that, for any Grothendieck topos $\mathcal{E}$, the functor

$$u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})} : \mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E}) \to \mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E})$$

of section 5.2.3 is faithful.

By Theorem 5.2.5, for any formula $\{\vec{x} \,.\, \phi\}$ presenting a $\mathbb{T}$-model $M_{\{\vec{x} \,.\, \phi\}}$ and any flat functor $F : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$,

$$u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})}(F)(\{\vec{x} \,.\, \phi\}) \cong F(M_{\{\vec{x} \,.\, \phi\}}).$$

Our thesis thus follows at once from Lemma 7.2.24. $\square$

7.2.4 *Further reformulations of condition (iii) of Theorem 6.3.1*

In this section we present other reformulations of condition (iii) of Theorem 6.3.1 which are applicable in a variety of different contexts.

Let us work in the context of a faithful interpretation of theories as in section 7.2. Consider the functors

$$u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})} : \mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E}) \to \mathbb{T}\text{-mod}(\mathcal{E})$$

and

$$u^{\mathbb{T}'}_{(\mathcal{H},\mathcal{E})} : \mathbf{Flat}(\mathcal{H}^{\mathrm{op}}, \mathcal{E}) \to \mathbb{T}'\text{-mod}(\mathcal{E})$$

of section 5.2.3.

Suppose that $\mathbb{T}'$ is of presheaf type and that $\mathcal{H}$ contains a category $\mathcal{P}$ whose Cauchy completion is the category of finitely presentable $\mathbb{T}'$-models (so that $\mathbb{T}'$ is classified by the topos $[\mathcal{P}, \mathbf{Set}]$). Notice that, since the functor $f : \mathcal{K} \to \mathcal{H}$ is a restriction of the functor $a_{\mathbf{Set}}$, $\mathcal{K}$ gets identified under it with a subcategory of $\mathcal{H}$ and hence, by Remark 5.2.2(c), the extension functor

$$\xi_{\mathcal{E}} : \mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E}) \to \mathbf{Flat}(\mathcal{H}^{\mathrm{op}}, \mathcal{E})$$

along the geometric morphism $[f, \mathbf{Set}]$ is faithful.

From the fact that $\mathbb{T}'$ faithfully interprets in $\mathbb{T}$ (in the sense of Definition 7.2.1) it follows that the functor $u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})}$ is faithful (resp. full and faithful) if and only if the composite functor $u^{\mathbb{T}'}_{(\mathcal{H},\mathcal{E})} \circ \xi_{\mathcal{E}}$ is, when regarded as a functor with values in the subcategory $\mathrm{Im}(a_{\mathcal{E}})$ of $\mathbb{T}'$-mod$(\mathcal{E})$. The interest of this reformulation lies in the alternative description of the functor

$$u^{\mathbb{T}'}_{(\mathcal{H},\mathcal{E})} : \mathbf{Flat}(\mathcal{H}^{\mathrm{op}}, \mathcal{E}) \to \mathbb{T}'\text{-mod}(\mathcal{E})$$

which is available under our hypotheses. Indeed, we have the following result:

Proposition 7.2.22 *Under the hypotheses specified above, the functor*

$$u^{\mathbb{T}'}_{(\mathcal{H},\mathcal{E})} : \mathbf{Flat}(\mathcal{H}^{\mathrm{op}}, \mathcal{E}) \to \mathbb{T}'\text{-mod}(\mathcal{E}) \simeq \mathbf{Flat}(\mathcal{P}^{\mathrm{op}}, \mathcal{E})$$

sends any flat functor $F : \mathcal{H}^{\mathrm{op}} \to \mathcal{E}$ to its restriction $F|_{\mathcal{P}^{\mathrm{op}}} : \mathcal{P}^{\mathrm{op}} \to \mathcal{E}$, and acts accordingly on the natural transformations.

Proof This follows immediately from Theorem 5.2.5 in view of the fact that $\mathcal{H}$ contains $\mathcal{P}$ and all the objects of $\mathcal{P}$ are finitely presentable $\mathbb{T}'$-models (since $\mathbb{T}'$ is by our hypotheses of presheaf type classified by the topos $[\mathcal{P}, \mathbf{Set}]$). □

As a corollary of the proposition, we immediately obtain the following result:

Theorem 7.2.23 *Under the hypotheses specified above*

(i) $\mathbb{T}$ *satisfies condition (iii)(b)-(1) of Remark 6.3.2(f) with respect to the category $\mathcal{K}$ if and only if, for any Grothendieck topos $\mathcal{E}$, every natural transformation between flat functors in the image of the functor $\xi_{\mathcal{E}}$ is determined by its restriction to the objects of $\mathcal{P}^{\mathrm{op}}$.*

(ii) $\mathbb{T}$ *satisfies condition (iii)(c) of Theorem 6.3.1 with respect to the category $\mathcal{K}$ if and only if, for any Grothendieck topos $\mathcal{E}$ and any $F, G \in \mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E})$, every natural transformation $\xi_{\mathcal{E}}(F)|_{\mathcal{P}^{\mathrm{op}}} \to \xi_{\mathcal{E}}(G)|_{\mathcal{P}^{\mathrm{op}}}$ in $\mathrm{Im}(a_{\mathcal{E}})$ is the image under $u^{\mathbb{T}'}_{(\mathcal{H},\mathcal{E})} \circ \xi_{\mathcal{E}}$ of a unique natural transformation $F \to G$.*

□

Theorem 7.2.23 can be notably applied for investigating the satisfaction of condition (iii)(b) of Theorem 6.3.1 by the injectivization of a geometric theory $\mathbb{T}$ satisfying the hypotheses of Proposition 7.2.18, as shown by the following proposition. Before stating it, we need the following lemma:

Lemma 7.2.24 *Let $\mathcal{C}$ be a small category, $\mathcal{E}$ a Grothendieck topos and $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{E}$ a flat functor. Then for any epimorphism $f : a \to b$ in $\mathcal{C}$ the arrow $F(f) : F(b) \to F(a)$ is monic in $\mathcal{E}$.*

Proof The thesis follows at once from the fact that flat functors preserve all the finite limits which exist in the domain category, using the well-known characterization of monomorphisms in terms of pullbacks. □

Proposition 7.2.25 *Let $\mathbb{T}$ be the injectivization (in the sense of Definition 7.2.10) of a theory of presheaf type $\mathbb{T}'$ whose finitely presentable models are all finitely generated (cf. for instance Proposition 7.2.18) and whose finitely generated models are all quotients (in the sense of section 7.2.3) of finitely presentable $\mathbb{T}'$-models (cf. for instance Proposition 7.2.19). Then $\mathbb{T}$ satisfies condition (iii)(b)-(1) of Remark 6.3.2(f) with respect to the category of finitely generated $\mathbb{T}'$-models.*

Proof It suffices to apply Lemma 7.2.24 in conjunction with Theorem 7.2.23 by taking $\mathcal{K}$ equal to the category of finitely generated $\mathbb{T}'$-models and sortwise injective homomorphisms between them, $\mathcal{H}$ equal to the category of finitely generated $\mathbb{T}'$-models and homomorphisms between them and $\mathcal{P}$ equal to the category of finitely presentable $\mathbb{T}'$-models and sortwise injective homomorphisms between them. □

We shall now establish a general result about injectivizations of theories of presheaf type, namely that for any theory of presheaf type $\mathbb{T}$ such that all the monic arrows in the category f.p.$\mathbb{T}$-mod(**Set**) are sortwise injective, the injectivization of $\mathbb{T}$ satisfies condition (iii)(c) of Theorem 6.3.1 with respect to the category of finitely presentable $\mathbb{T}$-models and sortwise injective homomorphisms between them.

Before proving this theorem, we need a number of preliminary results.

Lemma 7.2.26 *Let A be an object of a Grothendieck topos $\mathcal{E}$, $r : R \rightarrowtail A$ a subobject of A in $\mathcal{E}$, $e : E \to A$ an arrow in $\mathcal{E}$ and $\{f_i : E_i \to E \mid i \in I\}$ an epimorphic family in $\mathcal{E}$. If for every $i \in I$ the arrow $e \circ f_i$ factors through r then e factors through r.*

Proof The arrow $\coprod_{i\in I} f_i : \coprod_{i\in I} E_i \to E$ induced by the universal property of the coproduct is an epimorphism, since by our hypothesis the family $\{f_i : E_i \to E \mid i \in I\}$ is epimorphic. The factorizations $b_i : E_i \to R$ of the arrows $e \circ f_i$ through r induce an arrow $b := \coprod_{i\in I} b_i : \coprod_{i\in I} E_i \to R$ such that $r \circ b = e \circ \coprod_{i\in I} f_i$. Consider the epi-mono factorization $b = h \circ k$ in $\mathcal{E}$ of the arrow b, where $k : \coprod_{i\in I} E_i \twoheadrightarrow U$

and $h : U \rightarrowtail R$, and the epi-mono factorization $e = m \circ n$ of the arrow e in $\mathcal{E}$, where $n : E \twoheadrightarrow T$ and $m : T \rightarrowtail A$. Clearly, the arrow $e \circ \coprod_{i \in I} f_i$ factors both as $m \circ (n \circ \coprod_{i \in I} f_i)$ and as $(r \circ h) \circ k$. Now, as m and $r \circ h$ are monic and $n \circ \coprod_{i \in I} f_i$ and k are epic, the uniqueness of the epi-mono factorization of a given arrow in a topos implies that there exists an isomorphism $i : T \cong U$ such that $r \circ h \circ i = m$ and $i \circ n \circ \coprod_{i \in I} f_i = k$. The arrow $h \circ i \circ n$ thus provides a factorization of e through r, as required. □

The notation employed in the statements and proofs of the following results is borrowed from section 5.2.2.

The proof of the following lemma is straightforward from the results of section 5.2.2.

Lemma 7.2.27 *Let $F : \mathcal{D}^{\mathrm{op}} \to \mathcal{E}$ be a flat functor from a subcategory $\mathcal{D}$ of a small category $\mathcal{C}$ to a Grothendieck topos $\mathcal{E}$ and $x : E \to \tilde{F}(d)$ a generalized element which factors through $\chi_d^F : F(d) \to \tilde{F}(d)$ as $\chi_d^F \circ x'$. Then the natural transformation $\alpha_x : \gamma^*_{\mathcal{E}/E} \circ y_{\mathcal{C}}(d) \to !_E^* \circ \tilde{F}$ corresponding to x is equal to $\widetilde{\alpha_{x'}}$, where $\alpha_{x'}$ is the natural transformation $\gamma^*_{\mathcal{E}/E} \circ y_{\mathcal{D}}(d) \to !_E^* \circ F$ corresponding to x'.* □

Given a small category $\mathcal{C}$, we shall write $\mathcal{C}_m$ for the category whose objects are the objects of $\mathcal{C}$ and whose arrows are the monic arrows in $\mathcal{C}$ between them. The following results concern extensions $\tilde{F}$ of flat functors F along the embedding $\mathcal{C}_m \hookrightarrow \mathcal{C}$.

Proposition 7.2.28 *Let $\mathcal{C}$ be a small category and $F : (\mathcal{C}_m)^{\mathrm{op}} \to \mathcal{E}$ a flat functor. Then, for any object $c \in \mathcal{C}$ and any generalized element $x : E \to \tilde{F}(c)$, x factors through $\chi_c^F : F(c) \to \tilde{F}(c)$ if and only if the corresponding natural transformation $\alpha_x : \gamma^*_{\mathcal{E}/E} \circ y_{\mathcal{C}}(c) \to !_E^* \circ \tilde{F}$ is pointwise monic.*

Proof The 'only if' direction follows at once from Lemma 7.2.27 and Theorem 7.2.8. To prove the 'if' one, thanks to the localization technique, we can suppose without loss of generality that $E = 1_{\mathcal{E}}$.

Suppose that the natural transformation $\alpha_x : \gamma^*_{\mathcal{E}} \circ y_{\mathcal{C}}(c) \to \tilde{F}$ corresponding to the generalized element $x : 1_{\mathcal{E}} \to \tilde{F}(c)$ is pointwise monic. We want to prove that x factors through $\chi_c^F : F(c) \to \tilde{F}(c)$. Recall that for any object d of $\mathcal{C}$, $\alpha_x(d) : \coprod_{f \in \mathrm{Hom}_{\mathcal{C}}(d,c)} 1_{\mathcal{E}} \to \tilde{F}(d)$ is defined as the arrow which sends the coproduct arrow indexed by f to the generalized element $\tilde{F}(f) \circ x : 1_{\mathcal{E}} \to \tilde{F}(d)$.

By Lemma 7.2.12, if α_x is pointwise monic then, for any object $d \in \mathcal{C}$ and arrows $f, g : d \to c$ in $\mathcal{C}$, either $f = g$ or the equalizer of $\tilde{F}(f) \circ x$ and $\tilde{F}(g) \circ x$ is zero. Consider the pullbacks of the jointly epimorphic arrows $\kappa_{(a,z)} : F(a) \to \tilde{F}(c)$ along the arrow $x : 1_{\mathcal{E}} \to \tilde{F}(c)$:

$$\begin{tikzcd} E_{(a,z)} \arrow[r, "h_{(a,z)}"] \arrow[d, "e_{(a,z)}"'] & F(a) \arrow[d, "\kappa_{(a,z)}"] \\ 1_{\mathcal{E}} \arrow[r, "x"] & \tilde{F}(c). \end{tikzcd}$$

We shall prove that the epimorphic family $\{e_{(a,z)} : E_{(a,z)} \to 1_{\mathcal{E}} \mid (a,z) \in \mathcal{A}_c\}$ satisfies the condition that, for any $(a,z) \in \mathcal{A}_c$, the composite arrow $x \circ e_{(a,z)}$ factors through $\chi_c^F : F(c) \to \tilde{F}(c)$; this will imply our thesis by Lemma 7.2.26. As if $E_{(a,z)} \cong 0$ then $x \circ e_{(a,z)}$ factors through $\chi_c^F : F(c) \to \tilde{F}(c)$, it suffices to consider the case $E_{(a,z)} \not\cong 0$. Under this hypothesis, the arrow $z : c \to a$ is monic in $\mathcal{C}$. Indeed, for any two arrows $f, g : b \to c$ such that $z \circ f = z \circ g$, $\tilde{F}(f) \circ x \circ e_{(a,z)} = \tilde{F}(f) \circ \kappa_{(a,z)} \circ h_{(a,z)} = \tilde{F}(f) \circ \tilde{F}(z) \circ \chi_a^F \circ h_{(a,z)} = \tilde{F}(g) \circ \tilde{F}(z) \circ \chi_a^F \circ h_{(a,z)} = \tilde{F}(g) \circ \kappa_{(a,z)} \circ h_{(a,z)} = \tilde{F}(g) \circ x \circ e_{(a,z)}$; if the equalizer of $\tilde{F}(f) \circ x$ and $\tilde{F}(g) \circ x$ were zero then $E_{(a,z)}$ would also be isomorphic to zero (since it would admit an arrow to zero), so $f = g$. Now, z being monic, we have a factorization $\kappa_{(a,z)} = \chi_c^F \circ F(z)$ and hence $x \circ e_{(a,z)} = \kappa_{(a,z)} \circ h_{(a,z)}$ factors through χ_c^F, as required. □

Remark 7.2.29 By Proposition 7.2.14, Proposition 7.2.28 can be reformulated as follows: for any object $c \in \mathcal{C}$, $F(c) \cong \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}_{\mathcal{C}_m}\text{-mod}(\mathcal{E})}}(\gamma_{\mathcal{E}}^* \circ y_{\mathcal{C}}(c), \tilde{F})$, where $\mathbb{T}_{\mathcal{C}_m}$ is the geometric theory of flat functors on $(\mathcal{C}_m)^{\mathrm{op}}$.

For a small category $\mathcal{C}$ and a Grothendieck topos $\mathcal{E}$, let $\mathbf{Flat}_m(\mathcal{C}^{\mathrm{op}}, \mathcal{E})$ be the subcategory of $\mathbf{Flat}(\mathcal{C}^{\mathrm{op}}, \mathcal{E})$ whose objects are those of $\mathbf{Flat}(\mathcal{C}^{\mathrm{op}}, \mathcal{E})$ and whose arrows are the natural transformations between them which are pointwise monic.

Theorem 7.2.30 *Let $\mathcal{C}$ be a small category and $\mathcal{E}$ a Grothendieck topos. Then the extension functor*

$$\xi_{\mathcal{E}} : \mathbf{Flat}((\mathcal{C}_m)^{\mathrm{op}}, \mathcal{E}) \to \mathbf{Flat}(\mathcal{C}^{\mathrm{op}}, \mathcal{E})$$

along the embedding $\mathcal{C}_m \hookrightarrow \mathcal{C}$, which by Corollary 7.2.8 takes values into the subcategory $\mathbf{Flat}_m(\mathcal{C}^{\mathrm{op}}, \mathcal{E})$, is faithful and full on this latter category.

Proof We already know from Remark 5.2.2(c) that the functor $\xi_{\mathcal{E}}$ is faithful, so it remains to prove that it is full on $\mathbf{Flat}_m(\mathcal{C}^{\mathrm{op}}, \mathcal{E})$. Let F, G be flat functors $(\mathcal{C}_m)^{\mathrm{op}} \to \mathcal{E}$, and $\beta : \tilde{F} \to \tilde{G}$ be a pointwise monic natural transformation between them. We want to prove that there exists a natural transformation $\alpha : F \to G$ such that $\beta = \tilde{\alpha}$. It suffices to show that, for any $c \in \mathcal{C}$, $\beta(c) : \tilde{F}(c) \to \tilde{G}(c)$ restricts (along the arrows $\chi_c^F : F(c) \to \tilde{F}(c)$ and $\chi_c^G : G(c) \to \tilde{G}(c)$) to an arrow $F(c) \to G(c)$. To this end, we define a function $\gamma_E : \mathrm{Hom}_{\mathcal{E}}(E, F(c)) \to \mathrm{Hom}_{\mathcal{E}}(E, G(c))$ natural in $E \in \mathcal{E}$. By Remark 5.2.2(c), the set $\mathrm{Hom}_{\mathcal{E}}(E, F(c))$ (resp. the set $\mathrm{Hom}_{\mathcal{E}}(E, G(c))$) can be identified with the set of arrows $E \to \tilde{F}(c)$ (resp. with the set of arrows $E \to \tilde{G}(c)$) which factor through χ_c^F (resp. through χ_c^G). By Proposition 7.2.28, for any flat functor $H : \mathcal{C}^{\mathrm{op}} \to \mathcal{E}$, the arrows

$E \to \tilde{H}(c)$ which factor through χ_c^H correspond precisely to the pointwise monic natural transformations $\gamma^*_{\mathcal{E}/E} \circ y_{\mathcal{C}}(c) \to !_E^* \circ \tilde{H}$. Now, since β is pointwise monic, $!_E^*(\beta) : !_E^* \circ \tilde{F} \to !_E^* \circ \tilde{G}$ is also pointwise monic (since the functor $!_E^*$ preserves monomorphisms, it being the inverse image of a geometric morphism); hence any generalized element $E \to \tilde{F}(c)$ which factors through χ_c^F gives rise, by composition with $!_E^*(\beta)$ of the corresponding, pointwise monic, natural transformation $\gamma^*_{\mathcal{E}/E} \circ y_{\mathcal{C}}(c) \to !_E^* \circ \tilde{F}$, to a pointwise monic natural transformation $\gamma^*_{\mathcal{E}/E} \circ y_{\mathcal{C}}(c) \to !_E^* \circ \tilde{G}$, that is to a generalized element $E \to \tilde{G}(c)$ which factors through χ_c^G. But this generalized element is precisely $\beta_c \circ x$. So $\beta(c) : \tilde{F}(c) \to \tilde{G}(c)$ restricts to an arrow $F(c) \to G(c)$, as required. □

The following proposition identifies a class of theories of presheaf type $\mathbb{T}$ with the property that the monic arrows of the category f.p.$\mathbb{T}$-mod(**Set**) are sortwise injective.

Proposition 7.2.31 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ such that, for every sort A of Σ, the formula $\{x^A . \top\}$ presents a $\mathbb{T}$-model, and let $f : M \to N$ be a homomorphism of finitely presentable $\mathbb{T}$-models M and N. Then f is monic as an arrow of* f.p.$\mathbb{T}$-mod(**Set**) *(equivalently, as an arrow of* $\mathbb{T}$-mod(**Set**)*) if and only if it is sortwise injective.*

Proof Let A be a sort of Σ. As $\{x^A . \top\}$ presents a $\mathbb{T}$-model F_A, for any $\mathbb{T}$-model P in **Set** we have an equivalence $\mathrm{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(F_A, P) \cong PA$ natural in P. In particular, we have equivalences $\mathrm{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(F_A, M) \cong MA$ and $\mathrm{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(F_A, N) \cong NA$ under which the function

$$f \circ - : \mathrm{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(F_A, M) \to \mathrm{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(F_A, N)$$

corresponds to the function $f_A : MA \to NA$. Now, if f is monic then the function $f \circ - : \mathrm{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(F_A, M) \to \mathrm{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(F_A, N)$ is injective, equivalently $f_A : MA \to NA$ is injective, as required. □

Theorem 7.2.32 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ such that every monic arrow in* f.p.$\mathbb{T}$-mod(**Set**) *is sortwise injective. Then the injectivization of $\mathbb{T}$ satisfies condition (iii)(c) of Theorem 6.3.1 with respect to the category* f.p.$\mathbb{T}$-mod$(\mathbf{Set})_m$.

Proof Since every monic arrow in f.p.$\mathbb{T}$-mod(**Set**) is sortwise injective, the category f.p.$\mathbb{T}$-mod$(\mathbf{Set})_m$ is a subcategory both of the category $\mathbb{T}_m$-mod(**Set**) and of the category f.p.$\mathbb{T}$-mod(**Set**). So, by Proposition 7.2.7, the composite functor

$$\xi_{\mathcal{E}} : \mathbf{Flat}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})_m^{\mathrm{op}}, \mathcal{E}) \to \mathbf{Flat}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, \mathcal{E}) \simeq \mathbb{T}\text{-mod}(\mathcal{E})$$

takes values in the subcategory $\mathbb{T}_m$-mod$(\mathcal{E})$ of $\mathbb{T}$-mod$(\mathcal{E})$. Now, $\mathbb{T}_m$ satisfies condition (iii)(c) of Theorem 6.3.1 with respect to the category f.p.$\mathbb{T}$-mod$(\mathbf{Set})_m$ if and only if the functor $\xi_{\mathcal{E}}$ is faithful and full on $\mathbb{T}_m$-mod($\mathcal{E}$) (cf. the discussion preceding Proposition 7.2.22). But this follows from Theorem 7.2.30 in light of the fact that for any sortwise monic $\mathbb{T}$-model homomorphism $f : M \to N$ in $\mathcal{E}$, the natural transformation $\alpha_f : F_M \to F_N$ corresponding to it under the canonical Morita equivalence

$$\mathbf{Flat}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E}) \simeq \mathbb{T}\text{-mod}(\mathcal{E})$$

for $\mathbb{T}$ is pointwise monic. This latter fact can be proved as follows. For any object D of the category f.p.$\mathbb{T}$-mod(**Set**), the value of α_f at D can be identified with the arrow $[[\vec{x} \,.\, \phi]]_M \to [[\vec{x} \,.\, \phi]]_N$ canonically induced by f, where $\phi(\vec{x})$ is 'the' formula which presents the model D; so if f is sortwise monic then $\alpha_f(D)$ is monic in $\mathcal{E}$, it being the restriction of a monic arrow (namely $f_{A_1} \times \cdots \times f_{A_n}$, where $\vec{x} = (x^{A_1}, \ldots, x^{A_n})$) along two subobjects. □

We shall say that a geometric expansion $\mathbb{T}'$ of a geometric theory $\mathbb{T}$ is *fully faithful* if for every Grothendieck topos $\mathcal{E}$ the induced functor

$$(p_{\mathbb{T}}^{\mathbb{T}'})_{\mathcal{E}} : \mathbb{T}'\text{-mod}(\mathcal{E}) \to \mathbb{T}\text{-mod}(\mathcal{E})$$

is full and faithful.

Corollary 7.2.33 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ. If every monic arrow in* f.p.$\mathbb{T}$-mod(**Set**) *is sortwise injective then there is a fully faithful expansion of the injectivization of $\mathbb{T}$ which is classified by the topos* [f.p.$\mathbb{T}$-mod$(\mathbf{Set})_m$, **Set**].

Proof Consider the canonical geometric morphism

$$p_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})_m} : [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})_m, \mathbf{Set}] \to \mathbf{Set}[\mathbb{T}_m]$$

(cf. section 5.2.3). By Theorem 7.1.5, this morphism is, up to equivalence, of the form $p_{\mathbb{T}_m}^{\mathbb{T}'}$ for some expansion $\mathbb{T}'$ of $\mathbb{T}_m$.

By Theorem 7.2.32, the theory $\mathbb{T}_m$ satisfies condition (iii)(c) of Theorem 6.3.1 with respect to the category f.p.$\mathbb{T}$-mod$(\mathbf{Set})_m$, whence the functor

$$(p_{\mathbb{T}_m}^{\mathbb{T}'})_{\mathcal{E}} : \mathbb{T}'\text{-mod}(\mathcal{E}) \to \mathbb{T}_m\text{-mod}(\mathcal{E})$$

is full and faithful, i.e. $\mathbb{T}'$ is a fully faithful expansion of $\mathbb{T}_m$. □

Notice that if the finitely presentable $\mathbb{T}$-models coincide with the finitely presentable $\mathbb{T}_m$-models then the expansion provided by Corollary 7.2.33 is a presheaf completion of the theory $\mathbb{T}_m$. A simple example of a theory satisfying this condition (and the hypotheses of the corollary) is the theory $\mathbb{A}$ of commutative rings with unit; indeed, the finitely presented commutative rings with unit are precisely the finitely generated ones, that is the finitely presentable models of $\mathbb{A}_m$.

7.2.5 *A criterion for injectivizations*

In this section we shall establish a result providing a sufficient condition for the injectivization of a theory of presheaf type of a certain form to be of presheaf type too. Before stating it, we need some preliminaries.

The following definition gives a natural topos-theoretic generalization of the standard notion of congruence on a set-based structure.

Definition 7.2.34 Let Σ be a one-sorted first-order signature and M a Σ-structure in a Grothendieck topos $\mathcal{E}$. An equivalence relation $R \rightarrowtail M \times M$ on M in $\mathcal{E}$ is said to be a *congruence* if for any function symbol f of Σ of arity n we have a commutative diagram

$$\begin{tikzcd} R^n \arrow[r] \arrow[d] & (M \times M)^n \cong M^n \times M^n \arrow[d, "Mf \times Mf"] \\ R \arrow[r] & M \times M. \end{tikzcd}$$

Below, for a one-sorted signature Σ, by a Σ-structure epimorphism in a topos $\mathcal{E}$ we mean a Σ-structure homomorphism whose underlying arrow is an epimorphism in $\mathcal{E}$.

Proposition 7.2.35 *Let Σ be a one-sorted first-order signature and M a Σ-structure in a Grothendieck topos $\mathcal{E}$. For any congruence R on M, there exists a Σ-structure M/R in $\mathcal{E}$ whose underlying object is the quotient in $\mathcal{E}$ of M by the relation R, and a Σ-structure epimorphism $p_R : M \to M/R$ given by the canonical projection. Conversely, for any Σ-structure epimorphism $q : M \to N$, the kernel pair R_q of q is a congruence on M such that q is isomorphic to M/R_q.*

Proof The thesis follows immediately from the exactness properties of Grothendieck toposes relating epimorphisms and equivalence relations. □

Proposition 7.2.36 *Let $\mathbb{T}$ be a geometric theory over a one-sorted signature Σ and M a $\mathbb{T}$-model in a Grothendieck topos $\mathcal{E}$. If the axioms of $\mathbb{T}$ are all of the form $(\phi \vdash_{\vec{x}} \psi)$, where ϕ is a geometric formula not containing any conjunctions, then for any congruence R on M the Σ-structure M/R is a $\mathbb{T}$-model.*

Proof The thesis can be easily proved by induction on the structure of geometric formulae over Σ, using the fact that the action of the canonical projection homomorphism $p_R : M \to M/R$ on subobjects (of powers of M) preserves the top subobject, the natural order on subobjects, image factorizations and arbitrary unions. □

The following lemma shows that one can always perform image factorizations of homomorphisms of structures in regular categories.

Lemma 7.2.37 *Let Σ be a first-order signature and $\mathcal{C}$ a regular category. Then every Σ-structure homomorphism $f : M \to N$ in $\mathcal{C}$ can be factored as $h \circ g$, where $h : N' \rightarrowtail N$ is a Σ-substructure of N and $g : M \to N'$ is sortwise a cover.*

Proof First, we notice that in any regular category finite products of covers are covers; indeed, composites of covers are covers (this can be easily proved by using the definition of a cover as an arrow orthogonal to the class of monomorphisms), and the product of two covers $f \times g$, where $f : A \to B$ and $g : C \to D$, is equal to the composite $(1_B \times g) \circ (f \times 1_C)$ of two arrows which are pullbacks of covers and hence covers.

For every sort A of Σ, we set $N'A$ equal to $\mathrm{Im}(f_A)$, g_A equal to the canonical cover $MA \to \mathrm{Im}(f_A)$ and h_A equal to the canonical subobject $N'A \rightarrowtail NA$. For any function symbol $\xi : A_1 \cdots A_n \to B$ of Σ, we set $N'\xi$ equal to the restriction $N'A_1 \times \cdots \times N'A_n \to N'B$ of $N\xi : NA_1 \times \cdots \times NA_n \to NB$. This restriction actually exists (and is unique) since f is a Σ-structure homomorphism and $g_{A_1} \times \cdots \times g_{A_n}$ is a cover. For any relation symbol R of Σ of type $A_1 \cdots A_n$, we set $N'R$ equal to the intersection of NR with the canonical subobject $N'A_1 \times \cdots \times N'A_n \rightarrowtail NA_1 \times \cdots \times NA_n$. Clearly, $f = h \circ g$, g is sortwise a cover and h is a Σ-substructure embedding, as required. $\square$

The following result, giving an explicit characterization of decidable objects in terms of their generalized elements, was stated in [62] as Exercise VIII.8(a).

Lemma 7.2.38 *Let $\mathcal{E}$ be a cocomplete topos and A an object of $\mathcal{E}$. Then A is decidable if and only if for any generalized elements $x, y : E \to A$, there exists an epimorphic family (which can be taken consisting of just two elements) $\{e_i : E_i \to E \mid i \in I\}$ such that for any $i \in I$, either $x \circ e_i = y \circ e_i$ or the equalizer of $x \circ e_i$ and $y \circ e_i$ is zero.*

Proof Let us suppose that A is decidable. Let $d_A : D_A \rightarrowtail A \times A$ be the complement of the diagonal subobject $\delta_A : A \rightarrowtail A \times A$. Consider the pullback of $< x, y >: E \to A \times A$ along d_A:

$$\begin{tikzcd} E' \arrow[r, "u"] \arrow[d, "s"] & D_A \arrow[d, "d_A"] \\ E \arrow[r, "<x{,}y>"] & A \times A \end{tikzcd}$$

The equalizer $i : R \rightarrowtail E'$ of $x \circ s$ and $y \circ s$ is zero; indeed, by definition of D_A, the diagram

$$\begin{tikzcd} 0 \arrow[r] \arrow[d] & A \arrow[d, "\delta_A"] \\ D_A \arrow[r, "d_A"] & A \times A. \end{tikzcd}$$

is a pullback, and the arrows $z := x \circ s \circ i = y \circ s \circ i : R \to A$ and $u \circ i : R \to D_A$ satisfy the condition $\delta_A \circ z = d_A \circ u \circ i$.

Let us denote by $t : E'' \rightarrowtail E$ the pullback of the subobject $\delta_A : A \rightarrowtail A \times A$ along $< x, y >$. The arrows $s : E' \to E$ and $t : E'' \to E$ are jointly epimorphic, as they are pullbacks of arrows which are jointly epimorphic, namely δ_A and d_A.

Therefore, by setting $I = \{0, 1\}$, $E_0 = E'$, $E_1 = E''$, $e_0 : E_0 \to E$ equal to s and $e_1 : E_1 \to E$ equal to t, we have that the family $\{e_i : E_i \to E \mid i \in I\}$ satisfies the condition in the statement of the lemma.

Conversely, let us suppose that the condition in the statement of the lemma is satisfied. To prove that A is decidable, we have to show that the subobject $\delta_A \cup \neg\delta_A \rightarrowtail A \times A$ is an isomorphism, in other words that every subobject $< x, y >: E \rightarrowtail A \times A$ factors through it.

For any generalized elements $x', y' : E' \to A$, the arrow $< x', y' >: E' \to A \times A$ factors through $\neg\delta_A \rightarrowtail A \times A$ if and only if the equalizer of x' and y' is zero. Indeed, $< x', y' >$ factors through $\neg\delta_A$ if and only if its image does, and by definition of $\neg\delta_A$ this holds if and only if the pullback of it (or equivalently, of $< x', y' >$) along δ_A is zero, i.e. if and only if the equalizer of x' and y' is zero. Now, by our hypotheses, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ such that for every $i \in I$ either $x \circ e_i = y \circ e_i$ or the equalizer of $x \circ e_i$ and $y \circ e_i$ is zero. By Lemma 7.2.26, to prove that $< x, y >$ factors through $\delta_A \vee \neg\delta_A \rightarrowtail A \times A$ it suffices to show that for any $i \in I$, $< x \circ e_i, y \circ e_i >$ factors through $\delta_A \vee \neg\delta_A \rightarrowtail A \times A$. But by our hypothesis, for any given $i \in I$, either $x \circ e_i = y \circ e_i$ (which implies that $< x \circ e_i, y \circ e_i >$ factors through $\delta_A : A \rightarrowtail A \times A$) or the equalizer of $x \circ e_i$ and $y \circ e_i$ is zero (which implies, as we have just seen, that $< x \circ e_i, y \circ e_i >$ factors through $\neg\delta_A \rightarrowtail A \times A$); therefore, for every $i \in I$, $< x \circ e_i, y \circ e_i >$ factors through $\delta_A \cup \neg\delta_A \rightarrowtail A \times A$, as required. $\square$

Proposition 7.2.39 *Let Σ be a one-sorted signature, c a finite Σ-structure in* **Set**, *M a Σ-structure in a Grothendieck topos $\mathcal{E}$ whose underlying object is decidable and E an object of $\mathcal{E}$. Then, for any Σ-structure homomorphism $f : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a quotient map $q_i : c \to c_i$, where c_i is a finite Σ-structure, and a disjunctive Σ-structure homomorphism (in the sense of Lemma 7.2.16) $J_i : c_i \rightarrowtail \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ such that $J_i \circ q_i = \mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ f$ for all $i \in I$:*

$$\begin{tikzcd}
c \arrow[r, "f"] \arrow[d, "q_i"] & \mathrm{Hom}_{\mathcal{E}}(E, M) \arrow[d, "\mathrm{Hom}_{\mathcal{E}}(e_i, M)"] \\
c_i \arrow[r, "J_i"] & \mathrm{Hom}_{\mathcal{E}}(E_i, M).
\end{tikzcd}$$

Proof Let us suppose that c has n elements $x_1, \ldots, x_n$. We know from Lemma 7.2.38 that for any pair (i, j), where $i, j \in \{1, \ldots, n\}$, there exist arrows $e_{(i,j)} : E_{(i,j)} \to E$ and $e'_{(i,j)} : E'_{(i,j)} \to E$ in $\mathcal{E}$ such that $e_{(i,j)}$ and $e'_{(i,j)}$ are jointly epimorphic, $f(x_i) \circ e_{(i,j)} = f(x_j) \circ e_{(i,j)}$ and the equalizer of $f(x_i) \circ e'_{(i,j)}$ and $f(x_j) \circ e'_{(i,j)}$ is zero. The iterated fibred product of all these epimorphic families (indexed by the pairs (i, j) such that $i, j \in \{1, \ldots, n\}$) thus yields an epimorphic family $\{u_k : U_k \to E \mid k \in K\}$ with the property that for every $k \in K$ there exists a subset $S_k \subseteq \{1, \ldots, n\} \times \{1, \ldots, n\}$ such that, for every $(i, j) \in S_k$, $f(x_i) \circ u_k = f(x_j) \circ u_k$ and, for every $(i, j) \notin S_k$, the equalizer of $f(x_i) \circ u_k$ and

$f(x_j) \circ u_k$ is zero. For each $k \in K$, consider the quotient $q_k : c \to c_q$ of c by the congruence generated by the pairs of the form (x_i, x_j) for $(i, j) \in S_k$. It is easily seen that the Σ-structure homomorphism $\mathrm{Hom}_{\mathcal{E}}(e_k, M) \circ f$ factors through q_k and that the resulting factorization is a disjunctive Σ-structure homomorphism. We have thus found a set of data satisfying the condition in the statement of the proposition, as required. □

Proposition 7.2.40 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ. Suppose that the finitely presentable $\mathbb{T}$-models coincide with the finitely presentable $\mathbb{T}_m$-models and that the following condition is satisfied: for any Grothendieck topos $\mathcal{E}$, object E of $\mathcal{E}$ and Σ-structure homomorphism $x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, where c is a finitely presentable $\mathbb{T}$-model and M is a sortwise decidable $\mathbb{T}$-model, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model homomorphism $f_i : c \to c_i$ in* f.p.$\mathbb{T}$-mod(**Set**) *and a sortwise disjunctive Σ-structure homomorphism $x_i : c_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ such that $x_i \circ f_i = \mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ x$. Then the injectivization $\mathbb{T}_m$ of $\mathbb{T}$ satisfies conditions (i) and (ii) of Theorem 6.3.1 with respect to its category of finitely presentable models.*

Proof The proposition represents the particular case of Theorem 7.2.3 for the faithful interpretation of $\mathbb{T}$ into its injectivization, with $\mathcal{K}$ and $\mathcal{H}$ equal to the category of finitely presentable $\mathbb{T}$-models. The hypotheses of the theorem are satisfied by Remark 7.2.4. □

Corollary 7.2.41 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ. Suppose that the finitely presentable $\mathbb{T}$-models coincide with the finitely presentable $\mathbb{T}_m$-models, that the monic arrows of the category* f.p.$\mathbb{T}$-mod(**Set**) *are all sortwise injective, and that the following condition is satisfied: for any Grothendieck topos $\mathcal{E}$, object E of $\mathcal{E}$ and Σ-structure homomorphism $x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, where c is a finitely presentable $\mathbb{T}$-model and M is a sortwise decidable $\mathbb{T}$-model, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model homomorphism $f_i : c \to c_i$ in* f.p.$\mathbb{T}$-mod(**Set**) *and a sortwise disjunctive Σ-structure homomorphism $x_i : c_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ such that $x_i \circ f_i = \mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ x$. Then the injectivization $\mathbb{T}_m$ of $\mathbb{T}$ is of presheaf type.*

Proof By Theorem 7.2.32, $\mathbb{T}_m$ satisfies condition (iii) of Theorem 6.3.1 (since all the monic arrows in the category f.p.$\mathbb{T}$-mod(**Set**) are sortwise injective), while Proposition 7.2.40 ensures that it satisfies conditions (i) and (ii) of Theorem 6.3.1. We can thus conclude that $\mathbb{T}_m$ is of presheaf type. □

Remark 7.2.42 By Propositions 7.2.18 and 7.2.20, every geometric theory $\mathbb{T}$ whose finitely presentable models are precisely the finitely generated ones and whose axioms are all of the form $(\phi \vdash_{\vec{x}} \psi)$, where ψ is a quantifier-free geometric formula, satisfies the property that the finitely presentable $\mathbb{T}$-models coincide with the finitely presentable $\mathbb{T}_m$-models.

As a consequence of Corollary 7.2.41, we obtain the following result:

Corollary 7.2.43 *Let* $\mathbb{T}$ *be a geometric theory over a one-sorted signature* Σ *such that every quotient of a finitely presentable* $\mathbb{T}$*-model is a* $\mathbb{T}$*-model (e.g. a theory whose axioms are all of the form* $(\phi \vdash_{\vec{x}} \psi)$*, where* ϕ *does not contain any conjunctions – cf. Proposition 7.2.36). Suppose that the finitely presentable* $\mathbb{T}$*-models are exactly the finite* $\mathbb{T}$*-models and that all the monic arrows in the category* f.p.$\mathbb{T}$-mod(**Set**) *are injective functions. Then if* $\mathbb{T}$ *is of presheaf type,* $\mathbb{T}_m$ *is of presheaf type too.*

Proof Propositions 7.2.36 and 7.2.39 ensure that all the conditions of Corollary 7.2.41 are satisfied. Therefore $\mathbb{T}_m$ is of presheaf type, as required. □